

REVIEW

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Sodium intake for athletes before, during and after exercise: review and recommendations

Alan J. McCubbin^{1*} 

Abstract

Athletes training and competing in sports with a large body water turnover (e.g., team, endurance, and racquet sports) commonly focus on sodium containing foods, beverages, and supplements before, during and after exercise. Despite this, current guidelines for these scenarios are non-specific, primarily due to a lack of available evidence at the time of their writing. Recent evidence, plus a more detailed interpretation of the existing literature, may allow for more specific recommendations. This review examines the common rationales given by athletes, coaches, and practitioners for the planned intake of sodium on a daily basis, before, during, and after exercise. Currently there is no evidence that athletes require a greater dietary sodium intake day-to-day, due to regulation of sodium losses via the kidneys and sweat glands. Whether before, during or after exercise, evidence suggests that it is the relationship between sodium and water that influences health and performance outcomes, not sodium balance or total body sodium content in isolation. This relationship between water and sodium, given its influence on extracellular osmolality, influences thirst drive and fluid retention (i.e., overall body water balance), as well as the distribution of fluid between intracellular and extracellular compartments. Recommendations for sodium must therefore be made relative to changes in both sodium losses and fluid balance (i.e., fluid losses *and* intake), rather than planning based on sodium losses alone as many athletes, coaches, and practitioners have traditionally done. Common problems relating to sodium intake or balance (e.g., gastrointestinal issues and exercise-associated hyponatraemia) generally arise because of a mismatch between sodium and fluid balance. Questions remain about the form of sodium (i.e., sodium chloride or sodium citrate), and the delivery method (i.e., in solution, foods, or capsules), and the role, if any, of oral sensing of salty taste generally, or sodium specifically. Finally, some recommendations provided are based on limited evidence, extrapolations of research to specific scenarios, and/or theoretical modelling. Further research should address these gaps, as well as reproducing findings across a broader range of athletes (based on age, sex and athletic ability) and exercise types (e.g., weight-bearing exercise, intermittent exercise).

Keywords Salt, Electrolytes, Hydration, Hyperhydration, Hyponatraemia, Cramping, Sports drinks

Introduction

For athletes who experience a large energy expenditure and body water turnover during exercise (e.g., team sports, endurance sports, and racquet sports), a common focus of nutrition strategies centres around the main nutrients that athletes “use or lose” during exercise: carbohydrate, water, and sodium. While some debate persists around their nuances, quantifiable guidelines exist

*Correspondence:

Alan J. McCubbin
alan.mccubbin@monash.edu

¹ Department of Nutrition, Dietetics and Food, Monash University, Level 1, 264 Ferntree Gully Road, Notting Hill, Victoria 3168, Australia



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for both carbohydrate and fluid intake before, during, and after exercise [1–3]. Guidelines for sodium intake, however, are far less specific, owing to the lack of evidence available at the time of their writing. For example, the American College of Sports Medicine Position Statement on Nutrition for Athletic Performance [1] states that sodium should be replaced during exercise “... when large sweat sodium losses occur”, but stops short of quantifying how much sodium should be replaced, the goal of replacement, or how to know if enough sodium has been replaced. The National Athletic Trainers’ Association Position Statement on Fluid Replacement for the Physically Active provides similar advice [3]. In ultra-endurance sports, recommendations have specifically cautioned against overconsumption of sodium, and instead suggested sodium be consumed *ad libitum* in regular sports foods and beverages [4].

Despite significant work to refine and validate laboratory and field techniques for quantifying sweat sodium losses [5–9], and improve our understanding of factors influencing these losses [10–15], there remains surprisingly little research on the practical use of such knowledge to inform nutrition strategies for athletes. In the absence of specific recommendations, athletes have turned to other information sources for guidance on sodium replacement. These include fellow athletes, coaches, and marketing content from electrolyte supplement companies [16]. The result of this is that athletes often have well-entrenched beliefs around sodium that are clearly not based on evidence [16, 17]. In recent years, these beliefs appear to have intensified and expanded, including an increasingly held view that daily sodium supplementation is necessary to maintain euhydration, even in the absence of significant exercise [18].

This review therefore aims to examine the published evidence for dietary sodium intake and supplementation day-to-day, and specifically before, during and after exercise, by examining the rationales commonly given for use. The review also attempts, where possible, to provide quantifiable guidance for practitioners in each scenario based on existing evidence, including recent developments.

Daily sodium requirements for athletes

For several decades, public health messages have encouraged reduced dietary sodium intake, due to the well-established link with salt-sensitive hypertension in the general population [19, 20]. A meta-analysis of population surveys in Australian adults using 24-h urinary sodium excretion (24-h UNa), observed sodium intakes of 4040 mg·day⁻¹ for males and 2936 mg·day⁻¹ for females [21], well above those in national nutrition recommendations (typically < 2400 mg·day⁻¹) [22–24]. Few

studies have assessed the habitual sodium intake of athletes specifically, possibly because the reference method (24-h UNa) significantly underestimates sodium losses (and therefore estimated intake) when participants exercise and lose sodium through sweating [25]. Even in sedentary populations only ~90% of daily sodium intake is excreted in the urine, and accepted practice is to multiply 24-h UNa values by 110% to estimate sodium intake [21]. A single-day study of adolescent (12–15 years) male Japanese soccer players on days of no or lower intensity training found an estimated mean (standard deviation) sodium intake of 3880 (1200) mg·day⁻¹, or 85 mg·kg⁻¹·day⁻¹ [26], likely reflecting the overall Japanese population’s status among the world’s higher daily sodium intakes [27]. An Australian study assessed 24-h UNa in 16 adult male endurance runners, cyclists, and triathletes, over 3 consecutive days [13]. Applying the correction above, mean sodium intake was 3869 mg·day⁻¹ (range: 2151–6654 mg·day⁻¹) or 51 mg·kg⁻¹·day⁻¹ (range: 29–89 mg·kg⁻¹·day⁻¹). Unpublished data from 15 female endurance athletes, using the identical methodology, observed a lower absolute (3176 mg·day⁻¹, range: 1535–6153 mg·day⁻¹) but not relative (51 mg·kg⁻¹·day⁻¹, range: 25–94 mg·kg⁻¹·day⁻¹) sodium intake compared to males, still well above the daily recommended intake. These values may also underestimate sodium intake, because the requirement to minimise or abstain from exercise during this time likely reduced total food and fluid intake, both from main meals, and from planned intake before, during and after exercise.

Although many athletes believe that they have increased daily sodium requirements compared to sedentary individuals [16], the data presented suggests that they are likely already consuming considerably more sodium than required through their daily diet. Since sodium intake is positively correlated with energy intake [28], it is likely that most athletes unknowingly increase daily sodium intake as their training volume increases, without purposeful supplementation. Furthermore, evidence from over half a century ago [29–32], and more recently in athletes specifically [13, 33], demonstrates that both urinary and sweat sodium losses respond to changes in sodium intake to maintain sodium balance, likely regulated with a circaseptan (~7 day) rhythm [34]. The recent anecdotal trend of athletes, regardless of training volume or competitive level, using daily sodium supplementation, appears to be driven by a message that “water doesn’t hydrate you”. This message, propagated on social media and in marketing of related products, is likely an overly literal interpretation of the failure to retain large, excessive quantities of plain water when consumed rapidly and in large boluses. This phenomenon, where a bolus of plain water reduces plasma osmolality

(P_{Osm}) and triggers increased diuresis compared to a solution with a high sodium concentration, has been well described [35]. The presence of food or macronutrients in fluids however, slows the rate of gastric emptying and therefore the absorption rate of the fluid bolus, such that retention of plain water is indistinguishable from other beverages [35–37]. For individuals not significantly hypohydrated, sodium supplementation is unnecessary to achieve a state of euhydration. Instead, moderating the pattern of excessive water consumption between meals when already euhydrated, is a more cost-effective strategy which also aligns with public health messages to moderate sodium intake [38].

Before exercise

Many athletes firmly believe that increasing sodium intake prior to exercise is beneficial [16]. Anecdotally, this is also recommended by many sport nutrition practitioners, across a range of team, individual, and endurance sports. The justification given for these practices, however, varies widely. This section explores the common rationales given for targeted pre-exercise sodium intake, and the evidence for each (summarised at the end of this section in Table 1).

Rationale 1: Sodium intake increases ad libitum fluid intake, improving pre-exercise hydration status

Sodium is the largest single contributor (along with chloride) to extracellular osmolality and P_{Osm} [48]. An increase in P_{Osm} of only 2–3% is sufficient to increase thirst perception, and has been clearly demonstrated via intravenous saline infusion [49]. Yet studies of euhydrated individuals at rest have returned mixed findings for the effect of acute oral sodium intake on ad libitum fluid intake, possibly due to varying amounts of sodium consumed, and/or the presence of other food components. A large acute dose (60 mg·kg⁻¹) of sodium consumed in capsules increased ad libitum fluid intake by 562 mL (68%) in athletes instructed to drink only when thirsty, in the 2 h prior to exercise [39]. In contrast, a more modest sodium dose consumed as salted mixed nuts (18 mg·kg⁻¹ in men and 12 mg·kg⁻¹ in women) did not influence thirst or fluid intake over 2 h, compared to unsalted versions of the same food [50]. It is not clear however, if the failure to increase thirst and fluid intake in the latter study was truly due to the lower sodium dose, or delayed gastric emptying from the energy and fibre content of the nuts [50]. For athletes who frequently present hypohydrated prior to training or competition, added sodium in the hours before exercise, assuming adequate fluid is available, may help stimulate thirst and achieve euhydration in the timeframe required. The minimum dose required to achieve a worthwhile effect remains unclear, but a

logical approach would be to consume a sodium intake similar to that used for pre-exercise hyperhydration (see section below), scaled down to the volume of fluid the athlete would ideally consume. However, when provided in solution (as opposed to capsules or food), beverage palatability for most athletes declines above a sodium concentration of ~120 mg·100 mL⁻¹ when consumed as sodium chloride (NaCl) [40, 51], which may discourage ad libitum intake rather than increase it. If athletes are frequently presenting in a hypohydrated state prior to exercise, it is likely that neither fluid volume nor sodium content is top-of-mind for them. In some cases, this may be resolved by simply drawing attention to the issue, but in others a lack of ad libitum drinking may relate to distractions relating to both logistical factors (e.g., training shortly after waking, travel to a sporting venue, media commitments, other physical preparation tasks, etc.) or psychological factors (e.g., pre-competition anxiety). In these scenarios, manipulations in fluid availability and sodium concentration may only be successfully achieved with support (e.g., support staff for professional athletes, parents/guardians or other supporters for junior and development athletes).

Rationale 2: Increasing total body sodium in anticipation of losses during exercise

Total body sodium (TBNa⁺) content in humans is typically ~1 g·kg⁻¹ body mass (BM) [52]. This sodium is located mostly in the extracellular fluid, as part of bone mineral content, and in other osmotically-inactive stores such as skin and muscle tissue [53]. Many athletes believe they need to maximise TBNa⁺ by “loading up” on dietary sodium in anticipation of large losses during exercise, sometimes several days or even up to one week prior to competition [16]. From this author’s experience, this belief is frequently triggered by sweat composition testing, if an athlete is identified as having above-average sweat sodium losses. It is also a common belief that depletion of TBNa⁺ increases the risk of exercise associated muscle cramps (EAMC), or exercise associated hyponatraemia (EAH) [16]. Whether TBNa⁺ maintenance during exercise is necessary to achieve these outcomes is discussed in later sections, but multiple days of sodium loading appears unhelpful regardless. The kidneys are the main avenue for sodium excretion at rest, and adjustments in natriuresis occur within as little as two hours of oral sodium intake, increasing by eight hours after intake, before tapering off as required to maintain sodium balance [54]. The retention of increased sodium intake over multiple days is therefore minimal. In an intervention study with significant dietary control, an increase in sodium intake to 100 mg·kg⁻¹·d⁻¹ for 3 days (approximately double habitual intake), was accompanied

Table 1 Evidence summary and recommendations for sodium intake before exercise

Rationale	Supported by evidence?	Situations for use	Recommendation	Key references
Sodium intake increases ad libitum fluid intake, improving pre-exercise hydration status.	Yes	Athletes who frequently present in a hypohydrated state prior to training or competition, or do not habitually consume adequate fluid to achieve a euhydrated state.	A focus on increased sodium intake in the hours before exercise may help increase osmotic thirst drive and ad libitum drinking behaviour towards the desired volume. ~275–420 mg sodium per 100 mL of the desired fluid intake when consumed in capsules or salty food, or ~ 100–125 mg sodium per 100 mL when consumed as part of the fluid itself. Adjusting sodium content of beverages to the athlete's desired palatability may also encourage fluid intake, regardless of osmotic thirst drive NaCitrate may provide a more palatable option than NaCl.	[39, 40]
Sodium intake increases total body sodium in anticipation of losses during exercise.	Chronic sodium loading: No Acute sodium loading (≤ 8 h): Yes	Currently, the only evidence-based use of pre-exercise sodium loading is for encouraging voluntary fluid intake (see above) and/or hyperhydration (see below).	Refer to the principles below for sodium dose and timing recommendations.	[13]
Sodium intake improves fluid retention for achieving pre-exercise euhydration or hyperhydration.	Non-elite endurance athletes in non-weight-bearing sports (e.g. indoor or flat terrain outdoor cycling): Yes Weightbearing exercise (e.g. running or uphill cycling): Possibly no Elite athletes: Unknown	When anticipated fluid losses will be high (higher intensity and hot/humid environments) and it will be impractical or not possible due to gastrointestinal tolerance to adequately replace fluid during exercise.	~ 275–420 mg sodium per 100 mL of fluid consumed, in the ~4 h before exercise. The lower end of the range likely reduces the risk of gastrointestinal symptoms.	[41–44]
Sodium restriction reduces body water retention and acutely reduces body mass, for use in weight-category sports.	Lack of direct evidence in athletes. Current evidence is unclear.	As part of an acute weight making strategy for weight-category sports.	This strategy becomes increasingly important as fluid volume increases (i.e., $> 10 \text{ mL}\cdot\text{kg}^{-1}$), and when hyperhydration is desired (typically 20–25 $\text{mL}\cdot\text{kg}^{-1}$). Pre-exercise hyperhydration can be achieved with sodium or glycerol loading. NaCl, NaCitrate, or NaBicarb can all improve fluid retention, although the three have not been directly compared. NaCitrate may provide a more palatable option than NaCl. Sodium in solution may be slightly more efficacious than capsules or food, but risks poor athlete acceptance due to the palatability of very salty solutions.	[13, 45]
Pre-exercise sodium balance impacts sodium losses during exercise.	> 12 h: Possibly ≤ 12 h: No	N/A	No clear recommendation can be made at present due to the lack of evidence clearly showing an effect of reducing sodium from usual intake on body mass. Acute sodium loading is preferred to chronically increased intake if pre-exercise sodium loading is desired.	[13, 46, 47]

by a mean 24-h UNa of $79 \text{ mg}\cdot\text{kg}^{-1}\cdot\text{d}^{-1}$, suggesting at most a retention of $63 \text{ mg}\cdot\text{kg}^{-1}$ (21% retention) over three days [13]. In contrast a one-off, acute sodium load of $75 \text{ mg}\cdot\text{kg}^{-1}$ retained approximately $58 \text{ mg}\cdot\text{kg}^{-1}$ (77% retention) within the final 4 h before exercise [55]. On the whole, a large increase in sodium intake temporarily increases TBNa^+ , but can be achieved as (or more) effectively with an acute pre-exercise strategy than a multi-day increased sodium intake.

Rationale 3: Sodium intake improves fluid retention for achieving pre-exercise euhydration or hyperhydration

As already described, ingesting a large bolus of plain water reduces P_{Osm} , increasing diuresis. The addition of sodium, either separately or within the fluid, will prevent or limit the fall in P_{Osm} and lead to greater acute fluid retention [39, 41, 48]. While unnecessary for maintenance of daily euhydration, acute sodium loading can be useful in the final hours before exercise to ensure euhydration at the point of exercise commencement, while reducing the inconvenience of excessive urination in this time. Existing guidelines suggest a fluid intake of $5\text{--}10 \text{ mL}\cdot\text{kg}^{-1}$ (equivalent to $0.5\text{--}1\%$ BM) in the 2–4 h prior to exercise, with additional sodium to improve retention [1]. Comparison of beverages suggests that improved fluid retention (compared to plain water) over 2–3 h occurs in already euhydrated individuals at rest when sodium content exceeds $\sim 100 \text{ mg}\cdot 100 \text{ mL}^{-1}$, a concentration associated with oral rehydration solutions rather than traditional sports drinks [35]. Greater or longer fluid retention may require even greater sodium concentrations (see hyperhydration below), although this is associated with reduced beverage palatability, and may discourage voluntary consumption (discussed above). As described above, increased daily sodium intake may increase sweat sodium losses during subsequent exercise, as a normal response to maintain sodium balance. This is unlikely with acute sodium loading however, due to the time delay between a change in localised aldosterone concentration and changes in sweat gland function [56, 57]. Indeed, sodium loading as part of an acute ($< 4 \text{ h}$) strategy has not observed differences in sweat sodium concentration ($[\text{Na}^+]_{\text{sweat}}$) during subsequent exercise [46].

Using sodium to retain ingested fluids also presents an opportunity to increase total body water (TBW) above euhydrated levels in the short term ($< 8 \text{ h}$) (i.e., hyperhydration). Pre-exercise hyperhydration is a strategy sometimes employed when fluid losses during exercise are expected to be large (e.g., high absolute exercise intensity and/or hot and humid conditions), and the nature of the event, and/or the athlete's gastrointestinal tolerance to fluid, prevents them drinking adequately to

prevent detrimental effects of hypohydration [58]. This has included marathon and race walk events [59], but anecdotally, hyperhydration has also been utilised in professional team sports (e.g., rugby league and American football) where matches are held in hot and humid conditions. Because osmotically-active sodium is mostly located in the extracellular fluid, hyperhydration using sodium will primarily retain the additional TBW in the extracellular fluid, compared to glycerol which disperses evenly, therefore retaining fluid in both intracellular and extracellular compartments [60]. To date, no studies have directly compared hyperhydration with sodium or glycerol on performance outcomes, but evidence suggests it is TBW, not its distribution, that influences heart rate, body temperature, and perceived effort during exercise [61, 62].

The typical fluid volume used for hyperhydration in published research is $20\text{--}25 \text{ mL}\cdot\text{kg}^{-1}$ (i.e., $2\text{--}2.5\%$ BM), consumed 1–4 h before exercise [42, 43]. Most studies report the sodium intake in $\text{mg}\cdot\text{kg}^{-1}$, taken either dissolved in the fluid consumed, or in capsules. In reality, sodium in these studies is dosed relative to fluid volume, to maintain an overall sodium concentration of the combined fluids, tablets and/or capsules that approaches ($120\text{--}130 \text{ mmol}\cdot\text{L}^{-1}$ or $276\text{--}300 \text{ mg}\cdot 100 \text{ mL}^{-1}$), or in some cases exceeds ($145\text{--}180 \text{ mmol}\cdot\text{L}^{-1}$ or $333\text{--}414 \text{ mg}\cdot 100 \text{ mL}^{-1}$), normal plasma sodium concentration ($[\text{Na}^+]_{\text{plasma}}$: $135\text{--}145 \text{ mmol}\cdot\text{L}^{-1}$ or $311\text{--}333 \text{ mg}\cdot 100 \text{ mL}^{-1}$) [43]. For athletes who struggle to tolerate the fluid volume described in published research, a smaller volume (e.g., $10\text{--}20 \text{ mL}\cdot\text{kg}^{-1}$) could be used for a less aggressive hyperhydration strategy, and sodium intake adjusted to maintain the same overall concentration. Gastrointestinal symptoms are sometimes reported by athletes undertaking hyperhydration protocols [43]. This is most likely to occur due to osmotic fluid shifts into the gastrointestinal tract and is seen when sodium is excessively dosed (i.e., exceeds $\sim 145 \text{ mmol}\cdot\text{L}^{-1}$ or $333 \text{ mg}\cdot 100 \text{ mL}^{-1}$ of consumed fluid volume) [63].

The form of sodium (NaCl, sodium citrate (NaCitrate), or sodium bicarbonate (NaBicarb)) and the delivery method (in solution or capsules) has received limited research attention but is important both for athlete acceptance (due to fluid palatability), and efficacy in fluid retention. NaCl is the most comprehensively studied and is well established for its ability to achieve hyperhydration [43]. NaCitrate and NaBicarb have also been used in some hyperhydration research, and are potentially attractive options due their effect on buffering capacity, and (for NaCitrate only) improved palatability compared to NaCl when consumed in solution [44]. While both appear effective when given in capsules (NaBicarb: 39% fluid retained; NaCitrate: 32% fluid retained; plain

water: 20% retained) [44], to date no studies have directly compared their efficacy to NaCl. One study has compared NaCl given in solution or tablets, finding 59% of the ingested fluid ($30 \text{ mL} \cdot \text{kg}^{-1}$ fat free mass) was retained after 3 h with the solution, compared to 46% with tablets [41]. The authors speculated this difference may be due to the time taken for tablets to dissolve and release their contents, allowing water absorption ahead of sodium and triggering initial increased diuresis [41]. Fluid retention with NaCl tablets was still equivocal to glycerol in solution in this study, suggesting both tablets and capsules still present an efficacious, and potentially more acceptable strategy, for athletes who struggle with the palatability of highly concentrated sodium solutions [41].

The final question is whether sodium hyperhydration improves performance in relevant conditions. Two recent systematic reviews both concluded that on the whole, hyperhydration does improve endurance exercise performance in warm or hot conditions [42, 43], regardless of whether sodium or glycerol is used [42]. However, both papers noted that the overall effect was small, and lacks evidence for elite athletes, females, and weight-bearing exercise (e.g., running). Two recent studies have begun to address two of these limitations. In the first, hyperhydration was induced with NaCl capsules in 12 non-elite female cyclists across phases 1 and 4 of the menstrual cycle. Following a 75 min steady state ride at 60% $\text{VO}_{2\text{max}}$ in 34°C and 60% relative humidity (RH), hyperhydration improved time trial performance (200 kJ, taking ~20–25 min) compared to placebo in phase 4, and a non-significant 5% improvement in phase 1 [64]. In the second study, hyperhydration with glycerol and NaCl combined was induced in 13 endurance runners (12 male, 1 female), with fluid retention ~1.7% BM greater in the intervention vs control trial [65]. Despite this, hyperhydration failed to improve outdoor half-marathon running performance in temperate ambient conditions (26–27°C dry bulb, 24°C wet-bulb globe temperature) [65]. It is not clear if this finding indicates a lack of performance benefit from hyperhydration in weight-bearing exercise in general, that more “optimal” hydration status does not improve half-marathon running performance specifically [66], or that hyperhydration is unlikely to improve outdoor exercise performance in temperature environments [67]. Further research in hot conditions, different exercise durations/intensities, and in both laboratory and outdoor field environments, may help to make this distinction.

In summary, sodium loading improves fluid retention and achieves pre-exercise hyperhydration with doses of ~275–420 mg sodium per 100 mL of fluid. NaCl provided in solution may be slightly more efficacious than capsules, but it is unclear if the equivalent dose from

NaCitrate or NaBicarb will be as efficacious, leaving the choice of method to the athlete and practitioner based on practical considerations. While hyperhydration can improve non-weight bearing exercise performance in the heat in males and females, it is not yet clear if this strategy is effective for elite athletes, and may not be efficacious in weight-bearing forms of exercise, although further research is required to confirm this.

Rationale 4: Sodium restriction induces body water/body mass losses for acute weight manipulation

Taking the opposite approach to sodium loading for fluid retention, some athletes in weight-category sports, including weightlifting [68] and combat sports [69], are known to deliberately restrict sodium intake with the aim of reducing fluid retention, to acutely reduce TBW and therefore body mass. Such strategies even feature in practitioner guidelines for acute weight making in these sports [70]. While long-term manipulation of sodium intake does not appear to impact body mass [34], the effect of acute, severe sodium restriction is less clear. The research cited in the abovementioned guideline did not compare a low sodium diet to usual intake [71]. Instead, this study investigated five days of very high (~8050 $\text{mg} \cdot \text{d}^{-1}$) and very low (<500 $\text{mg} \cdot \text{d}^{-1}$) sodium intakes in individuals with hypertension [71]. The lack of comparison to participants' usual sodium intake means it is not possible to determine if the observed differences in BM were due to increased sodium from usual intake, decrease from usual intake, or a combination of both. Furthermore, the clinical population investigated would be expected to have different physiological handling of sodium to normotensive athletes [72, 73]. In a more recent study comparing three days restricted (1157 $\text{mg} \cdot \text{d}^{-1}$ or 15 $\text{mg} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$) vs usual (mean estimated from UNa: 3869 $\text{mg} \cdot \text{d}^{-1}$ or 51 $\text{mg} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$) sodium intake with ad libitum fluid intake in endurance athletes, BM was reduced by 0.6 kg (~0.8% BM) [13]. However, TBW measured via multi-frequency bioelectric impedance was not different between trials (both 44.1 L). It therefore remains unclear if sodium restriction represents a valid strategy to contribute to acute weight making. It was highlighted that sodium restriction, when combined with a water loading strategy immediately preceding fluid restriction (designed to induce greater diuresis during the fluid restricted period), may increase the risk of developing hyponatraemia [45]. Follow up research suggests that increased diuresis, along with no change in natriuresis, prevents any disturbances in blood sodium concentration [74].

Separately, the use of sodium restriction to reduce body mass prior to competition has been anecdotally reported in weight-sensitive sports (e.g., road cycling).

This mainly appears to occur in the post-exercise rehydration period between one stage and another in multi-day events; therefore, this approach will be discussed in the post-exercise section of this review.

During exercise

Replacement of sweat sodium losses during exercise is one aspect of sodium intake that has the most strongly held beliefs amongst athletes [16]. In the absence of specific guidelines, a range of sodium intake approaches are used, from arbitrary $\text{mg}\cdot\text{h}^{-1}$ quantities based on online posts or practitioner opinions, to variable targets based on the result of sweat composition and/or sodium loss testing. This section reviews the common rationales given for sodium replacement during exercise, and where possible, provides suggestions for how to approach this based on current evidence (summarised at the end of this section in Table 2). The process of quantifying $[\text{Na}^+]_{\text{sweat}}$ and sodium losses is beyond the scope of this review: readers are directed to a previously published, comprehensive review on the topic [75].

Rationale 1: A specific sodium concentration enhances gastrointestinal fluid and/or carbohydrate absorption

A common belief, historically promoted by some producers of electrolyte supplements, is that a specific beverage sodium concentration is optimal for gastrointestinal fluid and/or glucose absorption [76, 77]. Given the interactivity of sodium, water and glucose in intestinal absorption [78], these are plausible suggestions. Multiple studies have investigated the effect of beverage sodium concentration on water absorption, including highly controlled 'triple lumen' studies, where fluids of various composition are fed directly into different parts of the proximal small intestine using an enteral feeding tube, to eliminate variability due to gastric emptying [79]. These studies are limited to describing intestinal absorption at the proximal small intestine due to the nature of placing the tube. In any case, for athletes the effect of a solution on gastric emptying is highly relevant, and studies using oral intake of fluids with varying sodium concentration, and water absorption measured by the appearance of deuterated water in circulation, are more informative in practical terms [80, 81]. These studies do not support the notion that beverage sodium concentration should be chosen specifically for its ability to enhance or impair fluid absorption. Athletes therefore do not need to choose between a beverage with one sodium concentration for optimising gastrointestinal absorption, and another for replacing sodium according to other needs (see sections below).

The cotransport of glucose with water and sodium also raises the question of whether beverage sodium content

influences glucose uptake into circulation. While this has been less frequently investigated than water absorption in an exercise-specific context, an investigation using isotopically-labelled glucose found no difference in glucose absorption kinetics from a 10% w/v glucose solution ingested just prior to 30 min of stationary cycling at 65% $\text{VO}_{2\text{max}}$ when the solution contained 0, 58, or 115 $\text{mg}\cdot 100\text{ mL}^{-1}$ of sodium [82]. The authors suggested that it is still possible that higher sodium concentrations may improve intestinal glucose absorption, but noted that such concentrations ($>115\text{ mg}\cdot 100\text{ mL}^{-1}$) are not commonly consumed during exercise due to issues with beverage palatability [82].

Rationale 2: Sodium increases thirst drive and ad libitum fluid intake

Using sodium during exercise to encourage thirst drive and increase ad libitum fluid intake is a plausible suggestion, given the previously discussed effect at rest. Ultra-endurance sport nutrition guidelines have even warned against overly aggressive sodium intake during exercise, with the concern (based on evidence from retrospective case studies) that it could drive excessive fluid consumption, paradoxically leading to EAH [4]. It is important to differentiate between studies investigating the effect of sodium on thirst and fluid intake at rest and those conducted during exercise, because fluid replacement during exercise usually occurs while $[\text{Na}^+]_{\text{plasma}}$ and P_{Osm} are rising due to sweat losses, whereas sodium and fluid intake at rest usually occurs in the context of a stable or falling P_{Osm} as fluid is consumed.

While numerous studies have compared the effect of sodium-containing fluids to water on ad libitum fluid intake during exercise, the majority also included carbohydrate and/or flavouring in the intervention beverage only [51, 83–87]. More recent studies, using capsules to blind sodium intake and vary its dose independent of beverage flavour or carbohydrate content, have resulted in mixed findings. In a 72 km outdoor cycling time trial in cool (13.8°C , 63% RH) conditions, participants (5 male, 4 female) consumed either salt capsules providing $700\text{ mg}\cdot\text{h}^{-1}$ sodium or placebo capsules [88]. Ad libitum water intake was $160\text{ mL}\cdot\text{h}^{-1}$ (59%) greater when salt capsules were consumed. In an official Ironman 70.3 triathlon (22.5°C , 37% RH), participants were randomised to consume salt capsules ($559\text{ mg}\cdot\text{h}^{-1}$ sodium) or placebo capsules ($112\text{ mg}\cdot\text{h}^{-1}$ sodium due to sodium from other sports nutrition products), and ad libitum fluid intake was recorded via self-report after the race [89]. The group consuming salt capsules reportedly consumed $101\text{ mL}\cdot\text{h}^{-1}$ (37%) more fluid than the placebo group [89].

In contrast to these findings, one recent study investigated ad libitum fluid intake using salt capsules in

Table 2 Evidence summary and recommendations for sodium intake during exercise

Rationale given	Supported by evidence?	Situations for use	Recommendation	Key references
A specific sodium intake or concentration improves gastrointestinal fluid and/or glucose absorption.	No	N/A	N/A	[79, 81, 82]
Sodium intake increases thirst drive and ad libitum fluid intake during exercise	Evidence is mixed, finding relatively small differences, especially with a partially fixed fluid intake to allow consistent carbohydrate intake. Differences may exist based on sodium in solution or capsules.	For athletes who tend to consume inadequate fluid for optimal hydration status during exercise.	When exercise duration > 4 h and sweat rate < 1800 ml·h ⁻¹ : Replace sodium for stable [Na ⁺] _{plasma} (see below) based on desired fluid balance. Otherwise: Adjust sodium based on palatability and taste preferences that encourage optimal fluid intake.	[33, 88, 89]
Sodium intake helps to prevent or manage exercise associated muscle cramps (EAMC)	Some support, but unlikely to be the sole (or largest) contributor. Most likely to occur in conditions of chronic sodium deficiency and large water turnover (extremely rare).	When hypohydration is followed by a large intake of plain water.	No specific recommendation can be made at this time.	[92, 93]
Sodium intake helps to prevent or manage exercise associated hyponatraemia	Yes, when fluid replacement is > 70% but < 100% of losses. Fluid replacement > 100% is not recommended, and sodium will have minimal effect on the consequences of fluid overload.	When desired fluid replacement is > 70% of losses (mostly duration > 4 h and sweat rate < 1800 ml·h ⁻¹).	Refer to Fig. 3 and/or Fig. 2	[62, 104, 106, 110, 111]
Sodium intake improves exercise performance	No, unless it results in improved fluid balance (see ad libitum fluid intake above).	N/A – refer to ad libitum fluid intake above.	N/A	[88, 89, 117]

a laboratory setting, with warm ambient conditions (30°C, 32% RH), during 5 h treadmill running [33]. Sodium replacement was personalised based on prior sweat composition testing (100% sodium replacement (1217 mg·h⁻¹) vs 0% in placebo), and fluid intake consisted of both a fixed base volume (332 mL·h⁻¹ or 34% fluid loss replacement) to provide carbohydrate (1 g·kg·h⁻¹ in a 20% w/v solution), plus additional ad libitum plain water intake. This strategy of fixed fluid intake for carbohydrate, and ad libitum carbohydrate-free fluid is common for endurance athletes to ensure consistent carbohydrate intake, while adjusting fluid to changes in environmental conditions, pacing, and thirst across the event duration. Differences in ad libitum fluid intake were only 67 mL·h⁻¹ between trials, and not statistically significantly different. No evidence was found to support the concern [4] that aggressive sodium replacement (100%) causes excessive thirst drive or over-consumption of fluid during prolonged exercise [33].

In shorter duration and intermittent team sports, the effect of sodium intake on ad libitum fluid intake may not be as apparent, because practical factors relating to fluid availability, opportunities to drink, and gastrointestinal tolerance, may be more influential than osmotic thirst drive [90]. Sodium intake may still impact voluntary fluid intake due to effects on beverage palatability, with sodium concentrations > 100 mg·100 mL⁻¹ less preferred [40, 91]. It is important to note, however, that sodium content is only one of many variables that influence beverage palatability [83].

In summary, sodium intake may impact ad libitum fluid intake, either via osmotic thirst drive in longer duration exercise, or via beverage palatability in both shorter and longer duration exercise. The sodium quantity required to increase osmotic thirst drive is relative to fluid intake, and likely needs to increase [Na⁺]_{plasma} to a greater extent than plain water to be effectual (see later section for more details), while beverage palatability is most often achieved when added sodium (consumed as NaCl) does not exceed 100 mg·100 mL⁻¹.

Rationale 3: Sodium prevents and/or treats exercise associated muscle cramping

Of all the reasons athletes provide for consuming sodium during exercise, prevention of EAMC is the most frequently and strongly cited [16, 17]. The exact aetiology of EAMC remains elusive, but recent reviews on the topic suggest a complex syndrome, potentially with multiple overlapping causes, that may vary between people on the same day, or between days in the same person [92, 93]. In regards to sodium, while some observational studies have suggested sweat sodium losses contribute to the development of EAMC during exercise [94, 95], others

have not observed an effect of [Na⁺]_{plasma}, [Na⁺]_{sweat}, TBNa⁺ deficit, fluid or sodium intake [17, 96–99], or only observe these effects in select athlete groups [100]. In their recent review of the topic, Maughan & Shirreffs [92] noted that in early studies of labourers in the 1930's, a rare combination of very low 24-h UNa (possibly indicative of chronically low dietary sodium intake), combined with frequent and severe dehydration, may have given rise to hypovolaemic hyponatraemia, and predisposed workers to cramping. Saline, but not plain water for rehydration, reversed this effect [101]. Such large deficits of TBNa⁺ are extremely rare in modern athletes, but recent intervention studies, using electrically stimulated cramping models, have demonstrated potential effects of sodium intake to prevent a rapid fall in [Na⁺]_{plasma} and P_{Osm} (within < 60 min) and increased resistance to cramp induction when combined with purposeful eccentric exercise to induce muscle damage [102, 103]. Collectively, these findings suggest that while sodium may play some role in the prevention of EAMC, it may do so by balancing water intake and preventing rapid falls in [Na⁺]_{plasma} and P_{Osm}, or treating hypovolaemic hyponatraemia, a condition expected to be rare in modern athletic contexts. Regardless, modern theories of EAMC suggest any effect of sodium is only one of several risk factors, and failure to address others may reduce the efficacy of sodium intake for this purpose. Currently, no specific sodium recommendations can be made for the prevention of EAMC, beyond perhaps preventing a rapid decline in [Na⁺]_{plasma} or EAH (see below for details).

Rationale 4: Preventing a fall in blood sodium concentration, and onset of exercise associated hyponatraemia

The development of EAH, most commonly seen in endurance and ultra-endurance sports, is a potentially fatal condition which is defined, in accordance with its name, as a [Na⁺]_{plasma} below the normal clinical reference range (135–145 mmol·L⁻¹) during or after exercise [104]. Despite this definition, the concerning nature of EAH actually relates to the osmotic movement of excessive fluid into the intracellular space, causing cellular expansion, particularly in the lungs and brain [104]. Whether sodium intake can help prevent EAH, or at least reduce the impact of excessive fluid intake on [Na⁺]_{plasma} has been the subject of lively debate over the past two decades, with arguments made for [105–107] and against this approach [97, 108, 109]. One of the key arguments against sodium having a significant impact on [Na⁺]_{plasma} during exercise comes from Noakes' reanalysis of data collected in an extensive laboratory study by Baker et al. [62, 108]. In this study, participants completed multiple treadmill running trials, comparing four levels of fluid

replacement (resulting in a 4% BM Loss, 2% BM Loss, 0% BM Loss, and 2% BM gain), and three levels of sodium intake at each fluid replacement level (0 mg·100 mL⁻¹, 41 mg·100 mL⁻¹, and 69 mg·100 mL⁻¹). [Na⁺]_{serum} was measured before and after each run. While Noakes concluded fluid intake was by far the more influential factor for the change in [Na⁺]_{serum}, it must be highlighted that this conclusion was inevitable from a study design where fluid replacement varied from a large fluid deficit (4% BM loss) to excessive intake (2% BM gain), while sodium intake varied from none to only a modest concentration (69 mg·100 mL⁻¹), which is frequently more than doubled by athletes using commonly available sports nutrition products.

The study of Baker et al. [62] also validated the use of the Kurtz-Nguyen equation (Eq. 1) for predicting changes in serum or plasma [Na⁺] based on known changes in fluid, sodium and potassium balance during exercise [110]. This equation can also be rearranged (Eq. 2) to predict the sodium intake required to prevent a fall in [Na⁺]_{plasma} under different conditions of fluid balance, potassium balance, and [Na⁺]_{sweat} [111]. Modelling with this equation demonstrated that for athletes with a slightly above-average [Na⁺]_{sweat} (50 mmol·L⁻¹), a fall in [Na⁺]_{plasma} would be expected once an athlete replaces ≥70% of fluid losses, in the absence of sodium intake (Fig. 2). This suggests dilutional EAH is

theoretically possible even without a net positive fluid balance, and this level of fluid replacement would be required to prevent TBW deficits equivalent to ≥2% BM when total fluid loss is substantial during exercise (most likely ultra-endurance events) (Fig. 1) [111].

Equation 1. Calculating the predicted post-exercise [Na⁺]_{plasma} (in mmol·L⁻¹), given pre-exercise [Na⁺]_{plasma} and changes in fluid (TBW in L), sodium and potassium (in mmol) balance throughout exercise using the Kurtz-Nguyen Eq. [110]:

$$Post[Na^+]_{plasma} = \frac{[(Pre[Na^+]_{plasma} + 23.8) \times TBW_{pre}] + (1.03 \times \Delta E)}{TBW_{post}} - 23.8$$

where ΔE is the balance of sodium and potassium (expressed in mmol):

$$\Delta E = Na^+ + K^+ intake - Na^+ + K^+ losses (sweat \& urine)$$

Equation 2. Calculating the sodium intake (in mmol) required to maintain any given [Na⁺]_{plasma} (in mmol·L⁻¹), given pre-exercise [Na⁺]_{plasma}, changes in fluid balance (TBW in L), sodium and potassium losses (in mmol) and potassium intake (mmol) throughout exercise using the rearranged Kurtz-Nguyen Eq. [110, 111]:

$$Na^+ intake = \frac{[TBW_{post} \times (Post[Na^+]_{plasma} + 23.8)] - [TBW_{pre} \times (Pre[Na^+]_{plasma} + 23.8)]}{1.03} + (Na^+ loss + K^+ loss - K^+ intake)$$

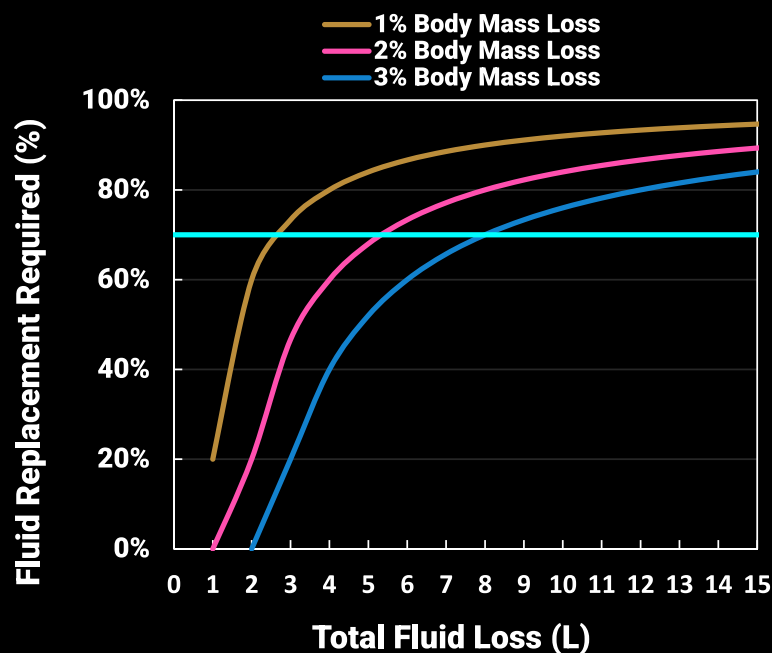


Fig. 1 Fluid replacement requirements (as a proportion of losses) to maintain a body water loss equivalent to 1%, 2% or 3% body mass during exercise. Notes: **a** Total fluid loss includes all sources of water loss, including sweat, urine, and respiratory losses. **b** Fluid replacement requirement includes all sources of body fluid gain, including exogenous (i.e., fluid intake), and endogenous (i.e., metabolic water production). **c** The red horizontal line (70% fluid replacement) indicates approximately where [Na⁺]_{plasma} would be expected to fall if whole-body [Na⁺]_{sweat} exceeds 50 mmol·L⁻¹ and no sodium is consumed

An argument has also been made that sodium replacement is unnecessary for preventing EAH, because osmotically-inactive sodium stores will be released in response to falling $[\text{Na}^+]_{\text{serum}}$ [112]. Evidence however, suggests that release of sodium stores occurs in response to hypovolaemia, not hyponatraemia [113–115]. Hyponatraemia is usually associated with hypervolaemia, and releasing additional sodium in this scenario would impair the ability to excrete the excess fluid. In contrast, releasing endogenous sodium in response to hypovolaemia may explain why hypovolaemic hyponatraemia, often described as a theoretical possibility, has not been convincingly demonstrated in athletes during exercise [104]. Because of the difficulties studying ultra-endurance exercise in a laboratory setting, and the challenges in accurately assessing fluid and sodium balance in field settings, it is unclear whether athletes maintain plasma volume (P_v) and contribute to maintenance of TBW due to their fluid and sodium intake practices, or if osmotically-inactive sodium stores are genuinely released to assist. It is also therefore unclear if purposeful sodium replacement, in scenarios where athletes replace >70% but <100% of fluid losses ($\geq 100\%$ fluid replacement is not recommended), is helpful in maintaining TBW and P_v for any given $[\text{Na}^+]_{\text{serum}}$ or P_{Osm} . Future research using biomarkers of sodium release, and/or MRI imaging of sodium stores [116], may provide novel insights and determine if use of the Kurtz-Nguyen equation is helpful to guide sodium replacement in such circumstances.

In summary, while excessive fluid intake rather than lack of sodium intake is the most common cause of EAH during exercise, sodium replacement still has some influence on $[\text{Na}^+]_{\text{plasma}}$, including when fluid is more completely (>70%) but not excessively ($\geq 100\%$) replaced. Mathematical modelling can help determine the % of sodium losses that theoretically need replacing to prevent a fall in $[\text{Na}^+]_{\text{plasma}}$ in these situations (Fig. 2), but there is currently a lack of evidence whether or not this either spares osmotically-inactive stores, or reduces the extent of hypovolaemia or TBW deficit.

Rationale 5: Sodium replacement improves exercise performance

Unlike other nutrients and non-nutrients (e.g., carbohydrate, fluid, caffeine), the effect of sodium replacement on performance has received little research attention, apart from the use of NaBicarb and NaCitrate in their capacity as buffering agents, which is beyond the scope of this review. A 2018 systematic review of sodium replacement and endurance performance identified only five studies that met the inclusion criteria, none of which were conducted in warm or hot ambient conditions (i.e., $\geq 30^\circ\text{C}$), nor personalised the sodium intake to the participant's losses [117]. Only one study identified a performance benefit from sodium replacement, which occurred during an actual Ironman 70.3 event, with participants randomised to consume salt or placebo capsules throughout the race [89]. Given the parallel group design, a large

% of fluid losses replaced										
	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%
Whole body $[\text{Na}^+]_{\text{sweat}}$										
20 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	0%	0%	0%	38%
30 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	0%	0%	6%	59%
40 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	0%	0%	29%	69%
50 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	0%	12%	43%	75%
60 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	0%	26%	53%	79%
70 mmol·L ⁻¹	0%	0%	0%	0%	0%	0%	14%	37%	60%	82%
80 mmol·L ⁻¹	0%	0%	0%	0%	0%	5%	25%	45%	65%	84%
% of sodium losses required to maintain stable $[\text{Na}^+]_{\text{plasma}}$										

Fig. 2 Calculated sodium replacement requirements, expressed as a proportion of sodium losses, for maintaining stable $[\text{Na}^+]_{\text{plasma}}$ during exercise. Data is presented from calculations showing differences in whole-body $[\text{Na}^+]_{\text{sweat}}$ (rows), and the proportion of sweat fluid losses replaced during exercise (columns). Assumes pre-exercise $[\text{Na}^+]_{\text{plasma}}$ of 140 mmol·L⁻¹, $[\text{K}^+]_{\text{sweat}}$ of 3.5 mmol·L⁻¹ (unreplaced), and no urine output. Calculations of a negative sodium requirement (i.e., $[\text{Na}^+]_{\text{plasma}}$ will rise without sodium intake) are defaulted to 0%. Data is based on calculations from [111]

performance difference (~8% faster in the sodium group) that exceeds well-established effects in other sport nutrition interventions on more controlled environments (e.g., carbohydrate, fluid or caffeine), and the potential influence of external factors (e.g., transition times, mechanical failures, drafting, wind exposure, etc.), it is difficult to express confidence in this finding, given no other study has observed a performance improvement from sodium consumption.

A small number of additional studies either did not meet the inclusion criteria for the abovementioned systematic review, or have been published since. The study of Baker et al. [62], which investigated four levels of fluid replacement and three sodium concentrations, included a treadmill running time-to-exhaustion (TTE) test at 85% $\dot{V}O_{2max}$ following the initial 2 h interval running protocol (7 × 15 min at 70% $\dot{V}O_{2max}$ with 2 min rest). TTE was improved in the 0% BM Loss and 2% BM gain trials compared to 2% and 4% BM loss, but within each level of fluid replacement, no effect of sodium intake was observed [62]. More recently, a randomised crossover study of beverage sodium concentration (0, 10, 20 and 50 mmol·L⁻¹) and tennis performance found that when players consumed 250 mL of the beverage before exercise, and 1000 mL throughout a 1-h training session, groundstroke accuracy (but not serve accuracy or agility test performance) was improved in the 50 mmol·L⁻¹ trial compared to placebo [118]. Unexpectedly, they also found that urine osmolality and perceived thirst were reduced as beverage sodium concentration increased.

The final study is, to date, the only study related to sodium replacement and performance that used sweat composition testing to personalise sodium replacement for each participant [119]. In this study, participants completed two 3-h cycling protocols at 90% of the first lactate threshold, (32°C, 50% RH), consuming fluid to replace either 60% or 80% of expected fluid losses. Sodium was replaced using the Kurtz-Nguyen equation to maintain $[Na^+]_{plasma}$ stability (99 mg·h⁻¹ in the 60% replacement trial, 390 mg·h⁻¹ in the 80% replacement trial). This was followed by a work-matched time trial, which was completed 8% faster in the 80% fluid replacement trial compared to 60% fluid replacement (1.9 vs 3.3% BM loss), confirming that it is the TBW deficit induced by hypo-hydration, rather than increased $[Na^+]_{plasma}$, that impairs endurance performance [62, 119].

While there is a clear lack of well-controlled studies of sodium replacement and exercise performance, collectively the existing studies suggest that with the exception of technique execution (i.e., tennis groundstroke accuracy), exercise performance appears to be improved by sodium intake only in scenarios where fluid intake is greater in the sodium than placebo or control trial. No

studies to date have shown a beneficial effect of sodium replacement on performance, independent of fluid replacement. Future studies should seek to address gaps in the existing literature in the following ways: a) conduct trials in a controlled, warm or hot environment (i.e., ≥ 30°C) where impacts of fluid and electrolyte losses are likely to be greatest; b) assess each participant's sodium losses during a prior visit (in identical conditions to experimental trials) to personalise sodium replacement in experimental trials; and c) investigate both fixed and ad libitum fluid intakes in separate studies, to distinguish any possible direct or indirect effects of sodium intake on performance.

How should we make sodium replacement recommendations during exercise?

Several potential reasons for sodium replacement during exercise have been explored in this section, with different ways of defining sodium intake. Traditionally, many practitioners, athletes and coaches have focussed on replacing sweat sodium losses (quantified through testing) to minimise TBNa⁺ deficit [16]. Intervention studies of prolonged, continuous exercise provide the simplest model for determining the efficacy of this approach. Most of these studies combined fixed, arbitrary (i.e., non-personalised) sodium replacement with a fixed fluid intake (often 100% replacement). Studies over 2 h [62], 4 h [61], and 6 h [120] induced sodium deficits of 3099 mg (~5% TBNa⁺), 4554 mg (6% TBNa⁺) and 4755 mg (6% TBNa⁺) respectively, finding no effect of sodium on heart rate, core body temperature, or perception of effort (RPE) during fixed intensity exercise. One recent study examined personalised, complete sodium replacement compared to no sodium during 5 h treadmill running, with ad libitum fluid intake [33]. Despite inducing a mean sodium deficit of 5658 mg (~8.5% TBNa⁺), the highest described in the literature to date, no effect on thirst, fluid intake or losses, heart rate, rectal temperature, or RPE, were observed. While larger sodium deficits could theoretically be incurred during ultra-endurance exercise with no sodium replacement, to date no studies (including observational studies from ultra-endurance events) have provided clear evidence that a certain level of sodium deficit during exercise can result in physiological, health, or performance consequences, independent of disturbances in fluid balance or distribution.

Given the evidence available to date, it is suggested that beyond palatability, specific sodium replacement during prolonged exercise is best planned based on the theoretical maintenance of $[Na^+]_{plasma}$ for a given, or potentially, desired fluid balance. Because $[Na^+]_{plasma}$ will generally rise during exercise unless > ~70% of fluid losses are replaced, the specific sporting scenarios where such an

approach could be warranted can be determined based on both the need, and practical limitations of achieving this level of fluid replacement. For practitioners, the following questions can be asked: 1) Would the athlete benefit from replacing >70% of their fluid losses in their training session or event? (refer to Eq. 3, acknowledging that some debate remains about the % BM loss that can be tolerated without performance impairment (i.e., acceptable BM loss), and whether this varies between weight-bearing and non-weight-bearing sports, or based on relative exercise intensity, athlete calibre, familiarisation with exercising in a hypohydrated state, and environmental conditions [67, 121–125]); 2) Can the athlete tolerate drinking enough fluid to replace >70% of their losses during their training session or event?; and 3) Is it practical, within the rules and limitations of the sport, for the athlete to replace >70% of their fluid losses during their training session or event? Only when the answer to all three questions is “yes” does there appear to be a case for such levels of fluid replacement, and therefore a theoretical need for sodium replacement beyond simply “seasoning to taste”. Typically, only endurance and ultra-endurance sports, or prolonged team sport training sessions in hot environments, would meet these criteria.

Equation 3. Calculating the % of fluid loss required to be replaced, where Acceptable BM loss is calculated as: (desired % BM loss from fluid/100) × BM (kg):

$$\text{Fluid required to be replaced (\% of losses)} = \frac{\text{Total sweat loss (L)} - \text{Acceptable BM loss (kg)}}{\text{Total sweat loss (L)}} \times 100$$

While many athletes may experience large sodium losses during exercise, the inability (or lack of need) to replace >70% of fluid losses means that $[\text{Na}^+]_{\text{plasma}}$ will inevitably rise even when no sodium is consumed, despite a large overall sodium deficit. While many athletes, coaches, and practitioners may feel anxious about not replacing such TBNa^+ losses, attempting to do so would exacerbate the rise in $[\text{Na}^+]_{\text{plasma}}$, in some cases to levels indicative of mild-to-moderate hypernatraemia. As an illustration of this point, the following data was captured from a 95 kg professional male rugby union player during an 80 min match in hot conditions: sweat rate of $2500 \text{ mL}\cdot\text{h}^{-1}$, $[\text{Na}^+]_{\text{sweat}}$ $60 \text{ mmol}\cdot\text{L}^{-1}$, and sweat sodium loss of $3450 \text{ mg}\cdot\text{h}^{-1}$. With limited opportunities, fluid intake is only 500 mL throughout the game (15% fluid replacement), losing 3.0% BM from fluid and 4600 mg of sodium ($\sim 4.8\%$ TBNa^+). If he began the match with a $[\text{Na}^+]_{\text{plasma}}$ of $140 \text{ mmol}\cdot\text{L}^{-1}$, his post-match $[\text{Na}^+]_{\text{plasma}}$ estimated from the Kurtz-Nguyen Eq. [110], would be $144 \text{ mmol}\cdot\text{L}^{-1}$ if no sodium was consumed. Attempting

to replace 100% of sodium losses (which some electrolyte supplement companies advocate) would result in an estimated post-match $[\text{Na}^+]_{\text{plasma}}$ of $148 \text{ mmol}\cdot\text{L}^{-1}$, above the clinical reference range. The scenario of large sodium deficit but increased $[\text{Na}^+]_{\text{plasma}}$ is more likely when sodium losses are mostly driven by high sweat rate rather than prolonged exercise duration. Such high sweat rates typically result from high absolute exercise intensities [126], which by their nature are time-limited due to fatigue. In contrast, ultra-endurance athletes can incur large sodium losses more as a function of exercise duration rather than sweat rate/exercise intensity. This scenario is less likely to result in a mismatch between sodium balance and $[\text{Na}^+]_{\text{plasma}}$, because the lower exercise intensity and sweat rate makes replacement of fluid losses more achievable (both the volume required and the gastrointestinal tolerance). Here, the greater % fluid replacement can be matched by sodium replacement, achieving both a reduction in overall sodium deficit, and maintenance of a stable $[\text{Na}^+]_{\text{plasma}}$.

The decision tree in Fig. 3 can be applied to determine when sweat composition testing and targeted sodium replacement is warranted, or when sodium replacement can be based on athlete taste preferences/palatability. If targeted replacement is warranted, practitioners can either utilise the rearranged Kurtz-Nguyen equation (Eq. 2), the outputs of that equation presented in Fig. 2,

or the summarised ranges presented in Fig. 3. When using these tools, fluid intake (or % replacement) can be based on the expected or desired fluid replacement for the scenario, keeping in mind individual variability in sweat rate, and personal and practical factors that may limit fluid replacement. By following this approach, an athlete who is under-consuming fluid for their needs will experience increased $[\text{Na}^+]_{\text{plasma}}$ which may facilitate thirst drive and encourage correction of fluid intake. Athletes over-consuming fluid would still experience either no change or a fall in $[\text{Na}^+]_{\text{plasma}}$, and therefore less thirst drive. The criteria in Fig. 3 also allows athletes and practitioners to quickly determine from the assessment of sweat fluid losses (which are simple and require minimal equipment) if there is any need to consider sweat composition testing for their athlete/s. In most scenarios outside of ultra-endurance sports, the figure demonstrates that sweat composition testing is unnecessary, and will not add value or help inform the athlete's sodium replacement strategy.

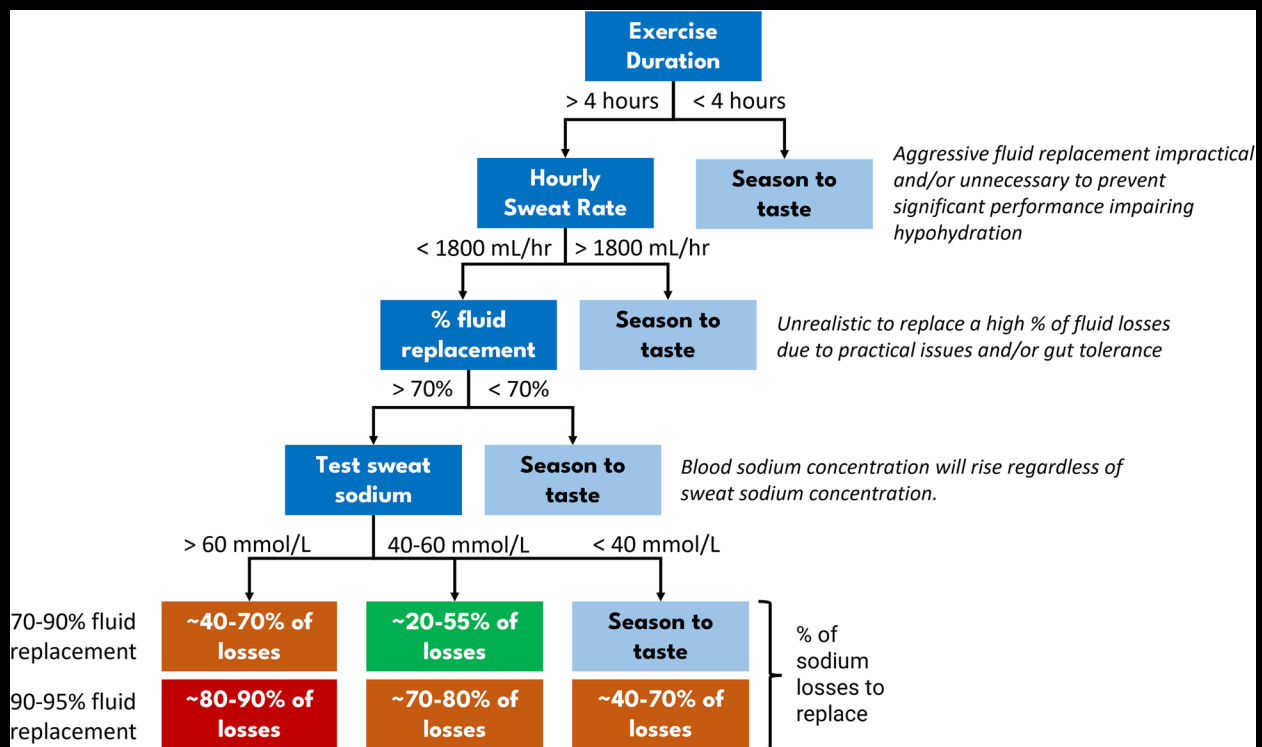


Fig. 3 Practical recommendations for sodium replacement during exercise, based on exercise duration, sweat rate, % fluid replacement, and whole-body sweat sodium concentration. In most scenarios other than ultra-endurance exercise, sodium replacement can be based on optimising food and fluid palatability (i.e., “season to taste”)

After exercise

As with before and during exercise, athletes generally believe that sodium intake after exercise, either to restore losses or to aid in rehydration strategies, is beneficial [16]. Since many of the rationales presented Here for post-exercise sodium intake are similar or the same as those discussed before exercise, this section will not re-evaluate the mechanisms Here, but instead focus on published evidence from studies investigating these rationales specifically in a post-exercise scenario. A summary of each rationale and recommendations are provided at the end of this section in Table 3.

Rationale 1: Sodium post-exercise increases thirst drive and ad libitum fluid intake

The effect of sodium for post-exercise rehydration has received significantly more research attention than either before or during exercise, spanning several decades. The vast majority of that research, however, has focussed on the retention of a prescribed fluid volume (discussed below). The utility of sodium to drive greater desire for fluid and voluntary fluid intake is important however, because a prescriptive rehydration plan is more likely to be effective when the athlete feels it is easy and/or enjoyable to follow. As with before exercise, there is a clear

theoretical rationale why increased sodium intake may drive greater ad libitum fluid intake in the hours following exercise. However, if the athlete requires rehydration post-exercise then by definition they are no longer in a euhydrated state, and will likely have a P_{Osm} and $[Na^+]_{plasma}$ that is greater than it was before exercise. If the athlete finishes exercise hypohydrated, the osmotic thirst drive should already be increased to some extent, with the larger the fluid deficit, the greater the effect [49]. It is therefore not immediately obvious if additional sodium is necessary to increase thirst drive post-exercise.

The post-exercise period is also confounded by other recovery goals, which often include carbohydrate intake for rapid glycogen resynthesis, and protein intake for muscle repair and adaptation [127]. It is therefore likely that a variety of foods and fluids will be consumed in the hours immediately following exercise, and it is the total sodium content from all foods, fluids and supplements, not just beverage composition, that will impact P_{Osm} , $[Na^+]_{plasma}$, and thirst drive. When food and fluid was provided ad libitum post-exercise in male participants but only a single beverage type allowed, mean fluid intake was 552 mL greater when consuming a carbohydrate-electrolyte beverage (sodium concentration 28 mg·100 mL⁻¹ and a 489 mg (32%) greater sodium

intake from food and fluids combined) compared to water [36]. However, this difference in fluid intake was not statistically significant, owing to substantial inter-individual variability. Furthermore, it cannot be determined from this study design whether any differences in fluid intake were driven by differences in sodium consumption, or other aspects of beverage palatability (e.g., sweetness, flavouring, participant beliefs about beverage suitability post-exercise, etc.). A similar study design in females yielded similar findings [37]. Further research, using either fluids in isolation, or combined with food sources of sodium that simulate common post-exercise recovery practices, may help determine any clear effect of sodium in this context.

Rationale 2: Sodium post-exercise increases fluid retention and post-exercise rehydration

The study of sodium intake and fluid retention during post-exercise rehydration is extensive, and spans several decades [35, 128–130]. Findings generally mirror those of pre-exercise fluid retention, whereby increased sodium intake (usually investigated as increased beverage sodium concentration) results in greater retention of a fixed fluid volume, following exercise that resulted in significant hypohydration (typically ~2% BM). The Beverage Hydration Index describes beverages not only based on variable sodium content, but also the presence of other nutrients frequently consumed in the post-exercise period [35, 130]. The concomitant consumption of these other nutrients, either in a beverage (e.g., milk) [35] or through accompanying foods [36, 37] also appears to improve fluid retention, when this outcome is defined based on body mass or net fluid balance (i.e., fluid intake vs urinary loss volume). The effect of other nutrients and the food matrix, however, may relate more to delayed gastric emptying, and therefore a slower delivery of fluid into circulation, rather than the reduction in osmotically-driven diuresis that occurs when fluids with a lower energy content are consumed together with sodium [37]. For this reason, the well-established role of sodium in fluid retention may be most impactful when lower energy and nutrient dense fluids are consumed in isolation post-exercise, with the aim of rapid and complete rehydration. The addition of sodium is likely less impactful when solid foods and/or nutrient dense beverages like milk are consumed as part of the athlete's overall recovery strategy.

Rationale 3: Sodium post-exercise restores whole body sodium content for the next exercise bout

The final rationale sometimes given is to ensure restoration of any TBNa^+ deficit incurred during exercise, similar to glycogen or fluid restoration. As discussed previously, urinary sodium excretion adjusts rapidly (within

1–2 h) of a disturbance in sodium balance. This impact post-exercise was first demonstrated in the 1950's in repeated ($n=48$) trials of a single participant [131]. More recently, a large (5658 mg or ~8.5% TBNa^+) sodium deficit in the placebo trial of a study already discussed [33], resulted in a 1035 mg (68%) reduction in 24-h UNa post-exercise compared to finishing exercise with no sodium deficit. This substantial sodium conservation by the kidneys occurred despite identical sodium intake in the post-exercise period, and was independent of urine volume [33]. Renal sodium conservation did not completely restore sodium balance within 24 h of such a large (5 h) exercise bout, but it demonstrates that complete post-exercise replacement of an exercise-induced sodium deficit is unnecessary. Another recent study, involving a smaller exercise-induced sodium deficit (~2500 mg) also observed evidence of renal sodium conservation post exercise, with a reduction in 24-h UNa of 1500 mg (45%) compared to a trial of the same sodium intake, when participants did not exercise [132]. Given the adjustment of natriuresis to varying sodium intakes and sweat losses, it is likely that urinary sodium conservation (and if exercise is performed in the coming days, sweat sodium conservation) will simply increase as necessary to restore medium-term sodium balance and TBNa^+ [13, 29, 32]. This effect is frequently observed in heat acclimation research [133–135]. Whether faster and more complete restoration of TBNa^+ is of any benefit for health or performance in subsequent exercise bouts that follow shortly after (i.e., within 24 h) is not clear, and requires additional research to determine.

Rationale 4: Sodium restriction post-exercise reduces restoration of TBW and therefore body mass prior to the following day's competition

As mentioned in the 'Before Exercise' section, there are anecdotal reports of athletes in multi-day, weight-sensitive sports (e.g., road cycling) deliberately restricting sodium in the post-exercise period, with the aim of reducing TBW restoration, and therefore reducing BM, prior to the following day's competition, which may include significant elevation gain. To the best of this author's knowledge, no studies have significantly reduced total dietary sodium intake for 12–24 h post-exercise, with either fixed or ad libitum fluid intake, and examined the effect on BM the following day. It has already been discussed that beverage sodium content does not influence fluid retention when food is consumed alongside fluid [36, 37]. Although these foods were not intentionally low in sodium content, a key mechanism explained for their efficacy appears to be reduced gastric emptying rate, rather than sodium intake itself. It is therefore unclear in a competition scenario like professional road

Table 3 Evidence summary and recommendations for sodium intake after exercise

Rationale given	Supported by evidence?	Situations for use	Recommendation	Key references
Sodium intake increases thirst drive and ad libitum fluid intake after exercise	Unclear due to lack of research using isolated sodium interventions and ad libitum fluid intake post-exercise. Plasma sodium and osmolality are already elevated due to hypohydration from exercise.	Possibly for athletes who do not habitually drink enough fluid to meet their rehydration goals.	A focus on increased sodium intake in the hours immediately after exercise may help increase osmotic thirst drive and ad libitum drinking behaviour towards the desired volume. ~275–420 mg sodium per 100 mL of the desired fluid intake when consumed in capsules or salty food, or ~100–125 mg sodium per 100 mL when consumed as part of the fluid itself. Adjusting sodium content of beverages to the athlete's desired palatability may also encourage fluid intake, regardless of osmotic thirst drive. NaCitrate may provide a more palatable option than NaCl when consuming solutions.	No key studies available to date
Sodium post-exercise increases fluid retention and post-exercise rehydration	When fluids of low energy/nutrient density are consumed in isolation post-exercise :- Yes. When fluids of greater energy/nutrient density (e.g. milk) or foods are consumed post-exercise: —No. Yes, but complete replacement is not required due to substantial renal sodium conservation after exercise. It is also not clear if there is any advantage of returning to a state of complete sodium balance prior to the next exercise bout.	When fluids of low energy/nutrient density are consumed in isolation post-exercise, and the main priority of recovery is rapid rehydration. Possibly when a large sodium deficit occurred during exercise that is similar to, or larger than, the athlete's typical daily sodium intake, and the recovery time to the next exercise bout is short (e.g., < 24 h).	Similar to pre-exercise hyperhydration (~275–420 mg sodium per 100 mL of fluid consumed), or possibly lower due to already elevated post-exercise plasma sodium and osmolality. No specific recommendation can be given at this time.	[35, 128–130]
Sodium post-exercise restores whole body sodium content for the next exercise bout	Currently a lack of direct evidence. Given the likely intake of energy dense foods in this scenario, and a recovery period of ≥ 12 h, fluid retention may not be impaired. Ad libitum fluid intake may be reduced by sodium restriction, but direct evidence is currently lacking. Evidence suggests that hypohydration may impair performance to a greater extent than reduced BM improves it.	In weight-sensitive or weight-bearing sports, where reduced BM is advantageous.	Increasing sodium intake from foods and beverages above the athlete's typical daily intake, is likely adequate to restore sodium balance within ~24 h if the sodium deficit during exercise was < ~2000 mg. For large sodium deficits, purposeful sodium replacement may aid the restoration of sodium balance. No specific recommendation can be given at this time.	[67, 136]
Sodium restriction post-exercise reduces fluid retention, allowing athletes to begin exercise the following day at a lower body mass			Evidence first needs to demonstrate advantages to hypohydration-induced reductions in BM on performance.	

cycling, where significant protein and carbohydrate intakes will be consumed as part of the recovery strategy, if reducing sodium intake dramatically will have a significant effect on retention of a fixed fluid volume during a recovery period of > 12 h.

Where sodium restriction may influence the rehydration process is in regard to thirst drive, and ad libitum fluid intake. While this has not been directly studied, the abovementioned study of sodium replacement during exercise provides a potential, indirect insight [33]. Here, sodium intake from food and fluids was fixed at 3174 mg in both trials during the post-exercise period, which concluded 24 h after exercise commenced the previous day. In the placebo trial, where athletes incurred a substantial (5750 mg) sodium deficit during exercise, urine volume in the post-exercise period was not different between trials. Ad libitum fluid intake was 543 mL lower however, resulting in a net fluid balance that was 812 mL lower than immediately prior to the previous day's exercise, compared with a positive fluid balance (+256 mL) when sodium was completely replaced during exercise.

While clear evidence of an effect of post-exercise sodium restriction is lacking, attention must be drawn to the potential negative consequences of attempting to do so. Given that any reduction in BM in this scenario will come from reduced TBW, this would come at a cost of beginning the following day's competition hypohydrated. The benefits or drawbacks of hypohydration on performance relative to BM (e.g., uphill cycling) is beyond the scope of this review, but readers are directed to studies that have either investigated this approach directly [136–140], or reported performance data in both absolute terms, and relative to BM [141], recognising potential pitfalls in interpreting hypohydration studies with participants unfamiliar with the exercise task, or the sensation of hypohydration itself [67]. Whether future research into post-exercise sodium restriction is warranted, is perhaps first predicated on evidence that reducing pre-exercise BM via hypohydration is advantageous to performance in the first place.

Unanswered questions and future research

While several gaps in knowledge have already been identified in this review, others are worthy of specific mention. Firstly, the differential effect of sodium ingested in solution vs tablets on thirst (outside of a fixed volume hyperhydration protocol [41]) has not been determined, nor has the effect on ad libitum fluid intake before, during and after exercise. Neural sensing of sodium in the oral cavity has been studied using a mouth rinsing protocol before, but only in relation to sprint exercise performance, with no beneficial effect observed [142]. A mouth rinsing protocol could be investigated in regard

to endurance or repeated sprint performance, and in the context of determining any effect on thirst and ad libitum fluid intake. Despite the growing popularity of NaCitrates in place of NaCl in carbohydrate-electrolyte drinks, no studies have compared their effects on gastrointestinal symptoms, changes in P_{Osm} and $[Na^+]_{plasma}$, thirst or ad libitum fluid intake during exercise, or post-exercise UNa.

Finally, many areas of this review and its associated recommendations are based on the results of a small number of studies, in some cases a single study, mostly in non-elite, male athletes. Some recommendations made in this review are also based on extrapolations of research to specific scenarios, and/or theoretical modelling. Given the pitfalls in reproducibility of research findings [143], and potentially divergent effects based on sex [144] and athletic ability [145], many areas discussed would benefit from additional study, especially across a broader range of participant types. Most of the studies in this review that examined the effect of sodium on outcomes during exercise were also conducted using continuous, submaximal, non-weight-bearing exercise protocols (primarily stationary cycling). Future research using weight-bearing exercise protocols (e.g., running, or uphill cycling), and intermittent exercise types that more closely resemble team and racquet sports (where large sodium losses are also possible), are warranted.

Conclusions

This review sought to examine the common rationales given by athletes for sodium intake on a day-to-day basis, as well as before, during, and after exercise. From this, the following conclusions are drawn:

- 1) There is no evidence that athletes need to supplement with sodium on a daily basis, outside of the specific exercise requirements listed below.
- 2) Acute sodium loading in the ~4 h prior to exercise can be used as part of a pre-exercise hyperhydration protocol, or to help drive thirst and ad libitum fluid intake in athletes who struggle to achieve euhydration prior to exercise. An intake of ~275–420 mg sodium per 100 mL fluid is most likely to maximise fluid intake and/or retention, but when consumed as NaCl in solution, may taste unpalatable and therefore limit the volume consumed. Strategies to overcome this include consuming NaCl in capsules or food (this may require additional time for gastric emptying and absorption), using NaCitrates instead of NaCl solutions (it is not yet clear if this is as efficacious for fluid retention), or reducing sodium concentration to ~100–120 mg·100 mL⁻¹ as a compromise between palatability and voluntary intake. There is no

evidence for a need to sodium load over an extended period (>4 h) prior to exercise, or that sodium loading will benefit athletes independent of stimulating greater fluid intake and retention.

- 3) Sodium intake during exercise may influence thirst and ad libitum fluid intake, but the importance of sodium type (NaCl or NaCitrate) or delivery method (solution or capsules) to this strategy is unclear. Replacing up to 100% of sodium losses does not appear to drive excessive thirst or fluid over-consumption. Sodium intake during exercise does not appear to influence performance unless it also results in a more optimal fluid intake. Replacement of sodium losses should therefore aim to complement fluid balance, rather than be seen as an independent or isolated strategy based solely on minimising reductions in TBNa⁺. This complementary strategy can include consuming sodium to prevent a fall in [Na⁺]_{plasma} during prolonged exercise, where replacement of >70%, but <100% of fluid losses may be both desirable and feasible. When athletes cannot or do not require >70% fluid replacement, [Na⁺]_{plasma} will rise during exercise, and sweat composition testing is unnecessary as it will not help inform the athlete's sodium replacement strategy.
- 4) Addition of sodium to post-exercise beverages can help retain fluid during rehydration when such beverages are consumed in isolation. When consumed together with food or nutrient-rich fluids (i.e., milk), sodium is unlikely to assist with restoring fluid balance. When a large sodium deficit has occurred during exercise, renal sodium conservation post-exercise means that complete sodium replacement is unnecessary to restore TBNa⁺. However, if rapid restoration is desirable (within 24 h), then increased dietary sodium intake may assist this process.

Finally, the above recommendations, and the evidence synthesised in their formulation, highlight the importance of the relationship between sodium and water balance in the body. Athletes, coaches, and practitioners should take from this review that sodium intake strategies will always be most effective when thought of as part of, and not independent from, hydration (fluid) strategies.

Abbreviations

24 h-UNa	24 Hour urinary sodium excretion
BM	Body mass
EAH	Exercise associated hyponatraemia
EAMC	Exercise associated muscle cramps
K ⁺	Potassium
[K ⁺]	Potassium concentration
Na ⁺	Sodium
[Na ⁺]	Sodium concentration
NaBicarb	Sodium bicarbonate
NaCitrate	Sodium citrate

NaCl	Sodium chloride
P _{Osm}	Plasma osmolality
P _v	Plasma volume
RH	Relative humidity
RPE	Rating of perceived exertion
TBNa ⁺	Total body sodium
TBW	Total body water
TTE	Time to exhaustion
UNa	Urinary sodium excretion
VO _{2max}	Maximal oxygen uptake

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References

1. Thomas DT, Erdman KA, Burke LM. American college of sports medicine joint position statement. Nutrition and athletic performance. *Med Sci Sports Exerc.* 2016;48(3):543–68.
2. Sawka MN, Burke LM, Eichner ER, Maughan RJ, Montain SJ, Stachenfeld NS. American college of sports medicine position stand. Exercise and fluid replacement. *Med Sci Sports Exerc.* 2007;39(2):377–90.
3. McDermott BP, Anderson SA, Armstrong LE, Casa DJ, Cheuvront SN, Cooper L, et al. National athletic trainers' association position statement: fluid replacement for the physically active. *J Athl Train.* 2017;52(9):877–95.
4. Hoffman MD, Stellingwerff T, Costa RJS. Considerations for ultra-endurance activities: part 2 - hydration. *Res Sports Med.* 2019;27(2):182–94.
5. Goulet EDB, Asselin A, Gosselin J, Baker LB. Measurement of sodium concentration in sweat samples: comparison of 5 analytical techniques. *Appl Physiol Nutr Metab.* 2017;42(8):861–8.
6. Goulet EDB, Baker LB. Sweat sodium concentration: inter-unit variability of a low cost, portable, and battery operated sodium analyzer. *Int J Sport Nutr Exerc Metab.* 2017;27(6):528–32.
7. Baker LB, Ungaro CT, Sopeña BC, Nuccio RP, Reimel AJ, Carter JM, et al. Body map of regional versus whole body sweating rate and sweat

- electrolyte concentrations in men and women during moderate exercise-heat stress. *J Appl Physiol.* 2018;124(5):1304–18.
8. Baker LB, Ungaro CT, Barnes KA, Nuccio RP, Reimel AJ, Stofan JR. Validity and reliability of a field technique for sweat Na⁺ and K⁺ analysis during exercise in a hot-humid environment. *Physiol Rep.* 2014;2(5):e12007.
 9. Barnes KA, Anderson ML, Stofan JR, Dalrymple KJ, Reimel AJ, Roberts TJ, et al. Normative data for sweating rate, sweat sodium concentration, and sweat sodium loss in athletes: an update and analysis by sport. *J Sports Sci.* 2019;37(20):2356–66.
 10. Baker LB, De Chavez PJD, Ungaro CT, Sopena BC, Nuccio RP, Reimel AJ, et al. Exercise intensity effects on total sweat electrolyte losses and regional vs. whole-body sweat [Na⁺], [Cl⁻], and [K⁺]. *Eur J Appl Physiol.* 2019;119(2):361–75.
 11. Buono MJ, Claros R, Deboer T, Wong J. Na⁺ secretion rate increases proportionally more than the Na⁺ reabsorption rate with increases in sweat rate. *J Appl Physiol.* 2008;105(4):1044–8.
 12. Buono MJ, Kolding M, Leslie E, Moreno D, Norwood S, Ordille A, et al. Heat acclimation causes a linear decrease in sweat sodium ion concentration. *J Therm Biol.* 2018;71:237–40.
 13. McCubbin AJ, Lopez MB, Cox GR, Caldwell Odgers JN, Costa RJS. Impact of 3-day high and low dietary sodium intake on sodium status in response to exertional-heat stress: a double-blind randomized control trial. *Eur J Appl Physiol.* 2019;119(9):2105–18.
 14. Baker LB, Ozga M, Merritt JR, Alfred S, De Chavez PJD, Hinkley JM. Mild dehydration does not alter acute changes in sweat electrolyte concentrations during exercise. *Physiol Rep.* 2024;12(18):e16174.
 15. Baker LB, De Chavez PJD, Nuccio RP, Brown SD, King MA, Sopena BC, et al. Explaining variation in sweat sodium concentration: effect of individual characteristics and exercise, environmental, and dietary factors. *J Appl Physiol.* 2022;133(6):1250–9.
 16. McCubbin AJ, Cox GR, Costa RJS. Sodium intake beliefs, information sources, and intended practices of endurance athletes before and during exercise. *Int J Sport Nutr Exerc Metab.* 2019;29(4):371–81.
 17. Hoffman MD, White MD. Belief in the need for sodium supplementation during ultramarathons remains strong: findings from the ultrarunners longitudinal TRacking (ULTRA) study. *Appl Physiol Nutr Metab.* 2020;45(2):118–22.
 18. Australian Broadcasting Corporation. Cooked: Electrolytes — who needs them?. (Science Friction). Available from: <https://www.abc.net.au/listen/programs/sciencefriction/electrolytes-and-hydration-science-sweat-and-sodium/104907782>. Cited 2025 Jun 22.
 19. Intersalt Cooperative Research Group. Intersalt: an international study of electrolyte excretion and blood pressure. Results for 24 hour urinary sodium and potassium excretion. *BMJ.* 1988;297(6644):319–28.
 20. Nishimoto M, Griffin KA, Wynne BM, Fujita T. Salt-sensitive hypertension and the kidney. *Hypertension.* 2024;81(6):1206–17.
 21. Land M, Neal BC, Johnson C, Nowson CA, Margerison C, Petersen KS. Salt consumption by Australian adults: a systematic review and meta-analysis. *Med J Aust.* 2018;208(2):75–81.
 22. Public Health England. Government recommendations for energy and nutrients for males and females aged 1 – 18 years and 19+ years. London; 2016 Jan. Available from: <https://www.gov.uk/government/publications/the-eatwell-guide>. Cited 2025 Jun 24.
 23. National Academies of Sciences, Engineering, and Medicine. Dietary Reference Intakes for sodium and potassium. Washington, DC: The National Academies Press; 2019 Jan. <https://doi.org/10.17226/25353>.
 24. National Health and Medical Research Council. Nutrient Reference Values for Australia and New Zealand Including Recommended Dietary Intakes. 2017 Jan. Available from: <https://www.eatforhealth.gov.au/nutrient-reference-values/nutrients/sodium>.
 25. McCubbin AJ. Exertional heat stress and sodium balance: leaders, followers, and adaptations. *Auton Neurosci.* 2021;235:102863.
 26. Nagashima Y, Horikawa A, Mori M. Association between 24 h urinary sodium and potassium excretion and dietary intake in Japanese male adolescent football players. *Adolescents.* 2021;1(4):461–72.
 27. Ikeda N, Yamaguchi M, Kashino I, Sugiyama T, Miura K, Nishi N. Evaluation of public health and economic impacts of dietary salt reduction initiatives on social security expenditures for cardiovascular disease control in Japan. *Hypertens Res.* 2025;48(4):1265–73.
 28. Murtaugh MA, Beasley JM, Appel LJ, Guenther PM, McFadden M, Greene T, et al. Relationship of sodium intake and blood pressure varies with energy intake: Secondary analysis of the DASH-Sodium trial. *Hypertens Dallas Tex.* 1979. 2018;71(5):858–65.
 29. McCance RA. The effect of salt deficiency in man on the volume of the extracellular fluids, and on the composition of sweat, saliva, gastric juice and cerebrospinal fluid. *J Physiol.* 1938;92(2):208–18.
 30. Robinson S, Nicholas JR, Smith JH, Daly WJ, Pearcy M. Time relation of renal and sweat gland adjustments to salt deficiency in men. *J Appl Physiol.* 1955;8(2):159–65.
 31. Robinson S, Kincaid RK, Rhamy RK. Effect of salt deficiency on the salt concentration in sweat. *J Appl Physiol.* 1950;3(2):55–62.
 32. Conn JW, Johnston LH, Louis LH. Relationship between salt intake and sweat salt concentration under conditions of hard work in humid heat. *Fed Proc.* 1946;5(1 Pt 2):230.
 33. McCubbin AJ, da Costa RJS. Effect of Personalized Sodium Replacement on Fluid and Sodium Balance and Thermophysiological Strain During and After Ultraendurance Running in the Heat. *Int J Sports Physiol Perform.* 2023;19(2):105–15.
 34. Rakova N, Juttner K, Dahlmann A, Schroder A, Linz P, Kopp C, et al. Long-term space flight simulation reveals infradian rhythmicity in human Na⁺ balance. *Cell Metab.* 2013;17(1):125–31.
 35. Maughan RJ, Watson P, Cordery PA, Walsh NP, Oliver SJ, Dolci A, et al. A randomized trial to assess the potential of different beverages to affect hydration status: development of a beverage hydration index. *Am J Clin Nutr.* 2016;103(3):717–23.
 36. Campagnolo N, Iudakhina E, Irwin C, Schubert M, Cox GR, Leveritt M, et al. Fluid, energy and nutrient recovery via ad libitum intake of different fluids and food. *Physiol Behav.* 2017;171:228–35.
 37. McCartney D, Irwin C, Cox GR, Desbrow B. Fluid, energy, and nutrient recovery via ad libitum intake of different commercial beverages and food in female athletes. *Appl Physiol Nutr Metab.* 2019;44(1):37–46.
 38. Gunn JP, Barron JL, Bowman BA, Merritt RK, Cogswell ME, Angell SY, et al. Sodium reduction is a public health priority: reflections on the institute of medicine's report, sodium intake in populations: assessment of evidence. *Am J Hypertens.* 2013;26(10):1178–80.
 39. Morris DM, Huot JR, Jetton AM, Collier SR, Utter AC. Acute sodium ingestion before exercise increases voluntary water consumption resulting in preexercise hyperhydration and improvement in exercise performance in the heat. *Int J Sport Nutr Exerc Metab.* 2015;25(5):456–62.
 40. Passe DH, Stofan JR, Rowe CL, Horswill CA, Murray R. Exercise condition affects hedonic responses to sodium in a sport drink. *Appetite.* 2009;52(3):561–7.
 41. Savoie FA, Asselin A, Goulet ED. Comparison of sodium chloride tablets-induced, sodium chloride solution-induced, and glycerol-induced hyperhydration on fluid balance responses in healthy men. *J Strength Cond Res.* 2016;30(10):2880–91.
 42. McCubbin AJ, Irwin C. The effect of pre-exercise oral hyperhydration on endurance exercise performance, heart rate and thermoregulation: a meta-analytical review. *Appl Physiol Nutr Metab.* 2024;49(5):569–83.
 43. Jardine WT, Aisbett B, Kelly MK, Burke LM, Ross ML, Condo D, et al. The effect of pre-exercise hyperhydration on exercise performance, physiological outcomes and gastrointestinal symptoms: a systematic review. *Sports Med.* 2023;53(11):2111–34.
 44. Siegler JC, Carr AJ, Jardine WT, Convit L, Cross R, Chapman D, et al. The hyperhydration potential of sodium bicarbonate and sodium citrate. *Int J Sport Nutr Exerc Metab.* 2022;32(2):74–81.
 45. Reale R, Slater G, Burke LM. Acute-weight-loss strategies for combat sports and applications to olympic success. *Int J Sports Physiol Perform.* 2017;12(2):142–51.
 46. Hamouti N, Fernandez-Elias VE, Ortega JF, Mora-Rodriguez R. Ingestion of sodium plus water improves cardiovascular function and performance during dehydrating cycling in the heat. *Scand J Med Sci Sports.* 2014;24(3):507–18.
 47. McCubbin AJ, Costa RJS. The impact of dietary sodium intake on sweat sodium concentration in response to endurance exercise: a systematic review. *Int J Sport Sci.* 2018;8(1):25–37.
 48. Armanini D, Bordin L, Dona' G, Andrisani A, Ambrosini G, Sabbadini C. Relationship between water and salt intake, osmolality, vasopressin, and aldosterone in the regulation of blood pressure. *J Clin Hypertens.* 2018;20(10):1455–7.

49. Stachenfeld NS. Acute effects of sodium ingestion on thirst and cardiovascular function. *Curr Sports Med Rep*. 2008;7(4 Suppl):S7–13.
50. Leshem M. Does salt increase thirst? *Appetite*. 2015;85:70–5.
51. Wemple RD, Morocco TS, Mack GW. Influence of sodium replacement on fluid ingestion following exercise-induced dehydration. *Int J Sport Nutr*. 1997;7(2):104–16.
52. Kennedy NSJ, Eastell R, Smith MA, Tothill P. Normal levels of total body sodium and chlorine by neutron activation analysis. *Phys Med Biol*. 1983;28(3):215–21.
53. Wiig H, Luft FC, Titze JM. The interstitium conducts extrarenal storage of sodium and represents a third compartment essential for extracellular volume and blood pressure homeostasis. *Acta Physiol*. 2018;222(3).
54. Carey RM. Evidence for a splanchnic sodium input monitor regulating renal sodium excretion in man. Lack of dependence upon aldosterone. *Circ Res*. 1978;43(1):19–23.
55. Goulet EDB, De La Flore A, Savoie FA, Gosselin J. Salt + Glycerol-induced hyperhydration enhances fluid retention more than salt- or glycerol-induced hyperhydration. *Int J Sport Nutr Exerc Metab*. 2018;28(3):246–52.
56. Collins KJ. The action of exogenous aldosterone on the secretion and composition of drug-induced sweat. *Clin Sci*. 1966;30(2):207–21.
57. Sato K, Dobson RL. The effect of intracutaneous d-aldosterone and hydrocortisone on human eccrine sweat gland function. *J Invest Dermatol*. 1970;54(6):450–62.
58. McCubbin AJ, Allanson BA, Caldwell Odgers JN, Cort MM, Costa RJS, Cox GR, et al. Sports dietitians Australia position statement: nutrition for exercise in hot environments. *Int J Sport Nutr Exerc Metab*. 2020;30(1):83–98.
59. Carr AJ, Vallance BS, Rothwell J, Rea AE, Burke LM, Guy JH. Competing in hot conditions at the Tokyo Olympic Games: preparation strategies used by Australian race walkers. *Front Physiol*. 2022;13:836858.
60. Nelson JL, Robergs RA. Exploring the potential ergogenic effects of glycerol hyperhydration. *Sports Med*. 2007;37(11):981–1000.
61. Sanders B, Noakes TD, Dennis SC. Sodium replacement and fluid shifts during prolonged exercise in humans. *Eur J Appl Physiol*. 2001;84(5):419–25.
62. Baker LB, Lang JA, Kenney WL. Quantitative analysis of serum sodium concentration after prolonged running in the heat. *J Appl Physiol*. 2008;105(1):91–9.
63. Sugihara A, Fujii N, Tsuji B, Watanabe K, Niwa T, Nishiyasu T. Hypervolemia induced by fluid ingestion at rest: effect of sodium concentration. *Eur J Appl Physiol*. 2014;114(10):2139–45.
64. Convit L, Orellana L, Périard JD, Carr AJ, Warmington S, Beaugeois M, et al. Sodium hyperhydration improves performance with no change in thermal and cardiovascular strain in female cyclists exercising in the heat across the menstrual cycle. *Int J Sport Nutr Exerc Metab*. 2024;1–13.
65. Jardine WT, Condo D, Aisbett B, Ross ML, Burke LM, Durkalec-Michalski K, et al. The addition of glycerol and sodium chloride to a hyperhydration protocol does not improve half-marathon time-trial performance in trained runners in warm conditions. *Int J Sports Physiol Perform*. 2025;20(6):770–8.
66. Lee MJ, Hammond KM, Vasdev A, Poole KL, Impey SG, Close GL, et al. Self-selecting fluid intake while maintaining high carbohydrate availability does not impair half-marathon performance. *Int J Sports Med*. 2014;35(14):1216–22.
67. James LJ, Funnell MP, James RM, Mears SA. Does hypohydration really impair endurance performance? methodological considerations for interpreting hydration research. *Sports Med*. 2019;49(Suppl 2):103–14.
68. Cox AM, Langan-Evans C, Jenkins D, Reale R, Pelly F, Slater GJ. Body mass management practices of olympic weightlifting athletes. *Int J Sport Nutr Exerc Metab*. 2025;35(1):67–75.
69. Reale R, Slater G, Burke LM. Weight management practices of australian olympic combat sport athletes. *Int J Sports Physiol Perform*. 2018;13(4):459–66.
70. Australian Institute of Sport. Making weight in weight category sports – Best practice guidelines for performance nutrition practitioners [Internet]. Canberra, ACT: Australian Institute of Sport; 2022 May p. 1–35. Available from: https://www.ausport.gov.au/ais/position_statements/best_practice_content/making-weight-in-weight-category-sports. Cited 2025 Aug 3.
71. He FJ, Markandu ND, Sagnella GA, MacGregor GA. Effect of salt intake on renal excretion of water in humans. *Hypertension*. 2001;38(3):317–20.
72. Titze J, Luft FC. Speculations on salt and the genesis of arterial hypertension. *Kidney Int*. 2017;91(6):1324–35.
73. Kolwelter J, Uder M, Schmieder RE. Tissue sodium content in hypertension and related organ damage. *J Hypertens*. 2020;38(12):2363–8.
74. Reale R, Slater G, Cox GR, Dunican IC, Burke LM. The effect of water loading on acute weight loss following fluid restriction in combat sports athletes. *Int J Sport Nutr Exerc Metab*. 2018;28(6):565–73.
75. Baker LB. Sweating rate and sweat sodium concentration in athletes: a review of methodology and intra/interindividual variability. *Sports Med*. 2017;47(Suppl 1):111–28.
76. Skratz Labs, Skratz Labs. What's Osmolality and Why It Matters for Your Sports Drink. Available from: <https://www.skratchlabs.com/blogs/science-products/whats-osmolality-and-why-it-matters>. Cited 2025 Jun 24.
77. Precision Hydration. Why sodium is crucial to athletes performing at their best?. Available from: <https://www.precisionhydration.com/performance-advice/hydration/sodium-for-athletes/>. Cited 2025 Jun 24.
78. Loo DDF, Zeuthen T, Chandy G, Wright EM. Cotransport of water by the Na⁺/glucose cotransporter. *Proc Natl Acad Sci*. 1996;93(23):13367–70.
79. Shi X, Passe DH. Water and solute absorption from carbohydrate-electrolyte solutions in the human proximal small intestine: a review and statistical analysis. *Int J Sport Nutr Exerc Metab*. 2010;20(5):427–42.
80. Gisolfi CV, Summers RD, Schedl HP, Bleiler TL. Effect of sodium concentration in a carbohydrate-electrolyte solution on intestinal absorption. *Med Sci Sports Exerc*. 1995;27(10):1414–20.
81. Jeukendrup AE, Currell K, Clarke J, Cole J, Blannin AK. Effect of beverage glucose and sodium content on fluid delivery. *Nutr Metab Lond*. 2009;6:9.
82. Hargreaves M, Costill D, Burke L, Mcconell G, Febbraio M. Influence of sodium on glucose bioavailability during exercise. *Med Sci Sports Exerc*. 1994;26(3):365.
83. Passe DH, Horn M, Murray R. Impact of beverage acceptability on fluid intake during exercise. *Appetite*. 2000;35(3):219–29.
84. Clapp AJ, Bishop PA, Smith JF, Mansfield ER. Effects of carbohydrate-electrolyte content of beverages on voluntary hydration in a simulated industrial environment. *AIHAJ*. 2000;61(5):692–9.
85. Baker LB, Munce TA, Kenney WL. Sex differences in voluntary fluid intake by older adults during exercise. *Med Sci Sports Exerc*. 2005;37(5):789–96.
86. Peacock OJ, Thompson D, Stokes KA. Voluntary drinking behaviour, fluid balance and psychological affect when ingesting water or a carbohydrate-electrolyte solution during exercise. *Appetite*. 2012;58(1):56–63.
87. Peacock OJ, Thompson D, Stokes KA. Impact of a carbohydrate-electrolyte drink on ingestive behaviour, affect and self-selected intensity during recreational exercise after 24-h fluid restriction. *Appetite*. 2013;60(1):5–12.
88. Cosgrove SD, Black KE. Sodium supplementation has no effect on endurance performance during a cycling time-trial in cool conditions: a randomised cross-over trial. *J Int Soc Sports Nutr*. 2013;10(1):30.
89. Del Coso J, Gonzalez-Millan C, Salinero JJ, Abian-Vicen J, Areces F, Lledo M, et al. Effects of oral salt supplementation on physical performance during a half-ironman: a randomized controlled trial. *Scand J Med Sci Sports*. 2016;26(2):156–64.
90. Belval LN, Hosokawa Y, Casa DJ, Adams WM, Armstrong LE, Baker LB, et al. Practical hydration solutions for sports. *Nutrients*. 2019;11(7):1550.
91. Kitson O, Rutherford-Markwick K, Fosskett A, Lee JKW, Diako C, Wong M, et al. Sensory perception of an oral rehydration solution during exercise in the heat. *Nutrients*. 2021;13(10):3313.
92. Maughan RJ, Shirreffs SM. Muscle cramping during exercise: causes, solutions, and questions remaining. *Sports Med*. 2019;49(Suppl 2):115–24.
93. Miller KC, McDermott BP, Yeargin SW, Fiol A, Schwellnus MP. An evidence-based review of the pathophysiology, treatment, and prevention of exercise-associated muscle cramps. *J Athl Train*. 2022;57(1):5–15.
94. Eichner ER. Genetic and other determinants of sweat sodium. *Curr Sports Med Rep*. 2008;7(Suppl. 1):S36–40.

95. Stofan JR, Zachwieja JJ, Horswill CA, Murray R, Anderson SA, Eichner ER. Sweat and sodium losses in NCAA football players: a precursor to heat cramps? *Int J Sport Nutr Exerc Metab.* 2005;15(6):641–52.
96. Sulzer NU, Schwellnus MP, Noakes TD. Serum electrolytes in Ironman triathletes with exercise-associated muscle cramping. *Med Sci Sports Exerc.* 2005;37(7):1081–5.
97. Hoffman MD, Stuempfle KJ, Valentino T. Sodium intake during an ultramarathon does not prevent muscle cramping, dehydration, hyponatremia, or nausea. *Sports Med.* 2015;1(1):39.
98. Hoffman MD, Stuempfle KJ. Muscle cramping during a 161-km ultramarathon: comparison of characteristics of those with and without cramping. *Sports Med.* 2015;1(1):24.
99. Hew-Butler TD, Sharwood K, Collins M, Speedy D, Noakes T. Sodium supplementation is not required to maintain serum sodium concentrations during an Ironman triathlon. *Br J Sports Med.* 2006;40(3):255–9.
100. Miller KC, McDermott BP, Yeargin SW. Sweat characteristics of cramp-prone and cramp-resistant athletes. *Int J Sport Nutr Exerc Metab.* 2020;30(3):218–28.
101. Dill DB, Bock AV, Edwards HT, Kennedy PH. Industrial fatigue. *J Ind Hyg Toxicol.* 1936;18:417–31.
102. Lau WY, Kato H, Nosaka K. Water intake after dehydration makes muscles more susceptible to cramp but electrolytes reverse that effect. *BMJ Open Sport Exerc Med.* 2019;5(1):e000478.
103. Lau WY, Kato H, Nosaka K. Effect of oral rehydration solution versus spring water intake during exercise in the heat on muscle cramp susceptibility of young men. *J Int Soc Sports Nutr.* 2021;18(1):22.
104. Hew-Butler T, Rosner MH, Fowkes-Godek S, Dugas JP, Hoffman MD, Lewis DP, et al. Statement of the 3rd international exercise-associated hyponatremia consensus development conference, Carlsbad, California, 2015. *Br J Sports Med.* 2015;49(22):1432–46.
105. Montain SJ, Sawka MN, Wenger CB. Hyponatremia associated with exercise: risk factors and pathogenesis. *Exerc Sport Sci Rev.* 2001;29(3):113–7.
106. Montain SJ, Chevront SN, Sawka MN. Exercise associated hyponatremia: quantitative analysis to understand the aetiology. *Br J Sports Med.* 2006;40(2):98–105; discussion 98–105.
107. Anastasiou CA, Kavouras SA, Arnaoutis G, Gioxari A, Kollia M, Botoula E, et al. Sodium replacement and plasma sodium drop during exercise in the heat when fluid intake matches fluid loss. *J Athl Train.* 2009;44(2):117–23.
108. Noakes TD. Changes in body mass alone explain almost all of the variance in the serum sodium concentrations during prolonged exercise. Has commercial influence impeded scientific endeavour? *Br J Sports Med.* 2011;45(6):475–7.
109. Hoffman MD, Stuempfle KJ. Sodium supplementation and exercise-associated hyponatremia during prolonged exercise. *Med Sci Sports Exerc.* 2015;47(9):1781–7.
110. Kurtz I, Nguyen MK. A simple quantitative approach to analyzing the generation of the dysnatremias. *Clin Exp Nephrol.* 2003;7(2):138–43.
111. McCubbin AJ. Modelling sodium requirements of athletes across a variety of exercise scenarios - identifying when to test and target, or season to taste. *Eur J Sport Sci.* 2023;23(6):992–1000.
112. Noakes TD, Sharwood K, Speedy D, Hew T, Reid S, Dugas J, et al. Three independent biological mechanisms cause exercise-associated hyponatremia: evidence from 2,135 weighed competitive athletic performances. *Proc Natl Acad Sci USA.* 2005;102(51):18550–5.
113. Nguyen MK, Nguyen DS, Nguyen MK. Osmotically inactive sodium and potassium storage: lessons learned from the Edelman and Boling data. *Am J Physiol-Ren Physiol.* 2016;311(3):F539–47.
114. Dahlmann A, Dorfelt K, Eicher F, Linz P, Kopp C, Mossinger I, et al. Magnetic resonance-determined sodium removal from tissue stores in hemodialysis patients. *Kidney Int.* 2015;87(2):434–41.
115. Hammon M, Grossmann S, Linz P, Kopp C, Dahlmann A, Garlisch C, et al. ²³Na magnetic resonance imaging of the lower leg of acute heart failure patients during diuretic treatment. *PLoS ONE.* 2015;10(10):e0141336.
116. Kopp C, Linz P, Wachsmuth L, Dahlmann A, Horbach T, Schoff C, et al. (23)Na magnetic resonance imaging of tissue sodium. *Hypertension.* 2012;59(1):167–72.
117. McCubbin AJ, Costa RJS. Impact of sodium ingestion during exercise on endurance performance: a systematic review. *Int J Sport Sci.* 2018;8(3):97–107.
118. Munson EH, Orange ST, Bray JW, Thurlow S, Marshall P, Vince RV. Sodium ingestion improves groundstroke performance in nationally-ranked tennis players: a randomized, placebo-controlled crossover trial. *Front Nutr.* 2020;7:549413.
119. Sage SA, Aldous JWF, Morris MG, Newell ML. Inadequate fluid replacement during prolonged cycling in the heat reduces performance independent of serum sodium concentrations. *Int J Sport Nutr Exerc Metab.* 2025;35(S1):S12.
120. Barr SJ, Costill DL, Fink WJ. Fluid replacement during prolonged exercise: effects of water, saline, or no fluid. *Med Sci Sports Exerc.* 1991;23(7):811–7.
121. Montain SJ, Sawka MN, Latzka WA, Valeri CR. Thermal and cardiovascular strain from hypohydration: influence of exercise intensity. *Int J Sports Med.* 1998;19(2):87–91.
122. Merry TL, Ainslie PN, Cotter JD. Effects of aerobic fitness on hypohydration-induced physiological strain and exercise impairment. *Acta Physiol.* 2010;198(2):179–90.
123. Deshayes TA, Daigle N, Jeker D, Lamontagne-Lacasse M, Perreault-Briere M, Claveau P, et al. Impact of repeated acute exposures to low and moderate exercise-induced hypohydration on physiological and subjective responses and endurance performance. *Nutrients.* 2021;13(12):4477.
124. Burke LM. Nutritional approaches to counter performance constraints in high-level sports competition. *Exp Physiol.* 2021;106(12):2304–23.
125. Deshayes TA, Pancrate T, Goulet EDB. Impact of dehydration on perceived exertion during endurance exercise: a systematic review with meta-analysis. *J Exerc Sci Fit.* 2022;20(3):224–35.
126. Gagnon D, Jay O, Kenny GP. The evaporative requirement for heat balance determines whole-body sweat rate during exercise under conditions permitting full evaporation. *J Physiol.* 2013;591(11):2925–35.
127. Russo I, Camões-Costa V, Gaskell SK, Porter J, Burke LM, Costa RJS. Systematic literature review: the effect of dairy milk on markers of recovery optimisation in response to endurance exercise. *Int J Sport Sci.* 2019;9(4):69–85.
128. Shirreffs SM, Maughan RJ. Volume repletion after exercise-induced volume depletion in humans: replacement of water and sodium losses. *Am J Physiol.* 1998;274(5):F868–875.
129. Merson SJ, Maughan RJ, Shirreffs SM. Rehydration with drinks differing in sodium concentration and recovery from moderate exercise-induced hypohydration in man. *Eur J Appl Physiol.* 2008;103(5):585–94.
130. Millard-Stafford M, Snow TK, Jones ML, Suh H. The beverage hydration index: influence of electrolytes, carbohydrate and protein. *Nutrients.* 2021;13(9).
131. Licht N. Competition between sweat glands and kidneys for salt and water in man. *J Appl Physiol.* 1957;11(2):223–6.
132. Macrae HZ, Cable T, Mouglin L, Nuttall H, Liddell E, Bellisario B, et al. Exercise-induced sweating decreases 24-h sodium balance compared to rest in recreational exercisers. *Proc Nutr Soc.* 2024;83(OCE4):E328.
133. Armstrong LE, Costill DL, Fink WJ, Bassett D, Hargreaves M, Nishibata I, et al. Effects of dietary sodium on body and muscle potassium content during heat acclimation. *Eur J Appl Physiol.* 1985;54(4):391–7.
134. Périard JD, Racinais S, Sawka MN. Adaptations and mechanisms of human heat acclimation: applications for competitive athletes and sports. *Scand J Med Sci Sports.* 2015;25(S1):20–38.
135. McDonald P, Brown HA, Topham TH, Kelly MK, Jardine WT, Carr A, et al. Influence of exercise heat acclimation protocol characteristics on adaptation kinetics: a quantitative review with Bayesian meta-regressions. *Compr Physiol.* 2025;15(3):e70017.
136. Ebert TR, Martin DT, Bullock N, Mujika I, Quod MJ, Farthing LA, et al. Influence of hydration status on thermoregulation and cycling hill climbing. *Med Sci Sports Exerc.* 2007;39(2):323.
137. Bardis CN, Kavouras SA, Arnaoutis G, Panagiotakos DB, Sidossis LS. Mild dehydration and cycling performance during 5-kilometer hill climbing. *J Athl Train.* 2013;48(6):741–7.
138. Bardis CN, Kavouras SA, Adams JD, Geladas ND, Panagiotakos DB, Sidossis LS. Prescribed drinking leads to better cycling performance than ad libitum drinking. *Med Sci Sports Exerc.* 2017;49(6):1244–51.

139. Adams JD, Sekiguchi Y, Suh HG, Seal AD, Sprong CA, Kirkland TW, et al. Dehydration impairs cycling performance, independently of thirst: a blinded study. *Med Sci Sports Exerc.* 2018;50(8):1697–703.
140. Adams JD, Scott DM, Brand NA, Suh HG, Seal AD, McDermott BP, et al. Mild hypohydration impairs cycle ergometry performance in the heat: a blinded study. *Scand J Med Sci Sports.* 2019;29(5):686–95.
141. Jeker D, Claveau P, Abed MEF, Deshayes TA, Lajoie C, Gendron P, et al. Programmed vs. Thirst-driven drinking during prolonged cycling in a warm environment. *Nutrients.* 2021;14(1).
142. Gray EA, Cavaleri R, Siegler JC. Mouth rinsing and ingestion of unpleasant salty or bitter solutions does not improve cycling sprint performance in trained cyclists. *Int J Sport Nutr Exerc Metab.* 2023;33(6):316–22.
143. Murphy J, Caldwell AR, Mesquida C, Ladell AJM, Encarnación-Martínez A, Tual A, et al. Estimating the replicability of sports and exercise science research. *Sports Med.* 2025. <https://doi.org/10.1007/s40279-025-02201-w>.
144. Smith ES, McKay AKA, Kuikman M, Ackerman KE, Harris R, Elliott-Sale KJ, et al. Auditing the representation of female versus male athletes in sports science and sports medicine research: evidence-based performance supplements. *Nutrients.* 2022;14(5):953.
145. McKay AKA, Stellingwerff T, Smith ES, Martin DT, Mujika I, Goosey-Tolfrey VL, et al. Defining training and performance caliber: a participant classification framework. *Int J Sports Physiol Perform.* 2022;17(2):317–31.

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